

Software Engineering

How To Create Software Products Everyone Loves!

Module 7

UI/UX Fundamentals: Screens, Layouts, Objects

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Screens

2

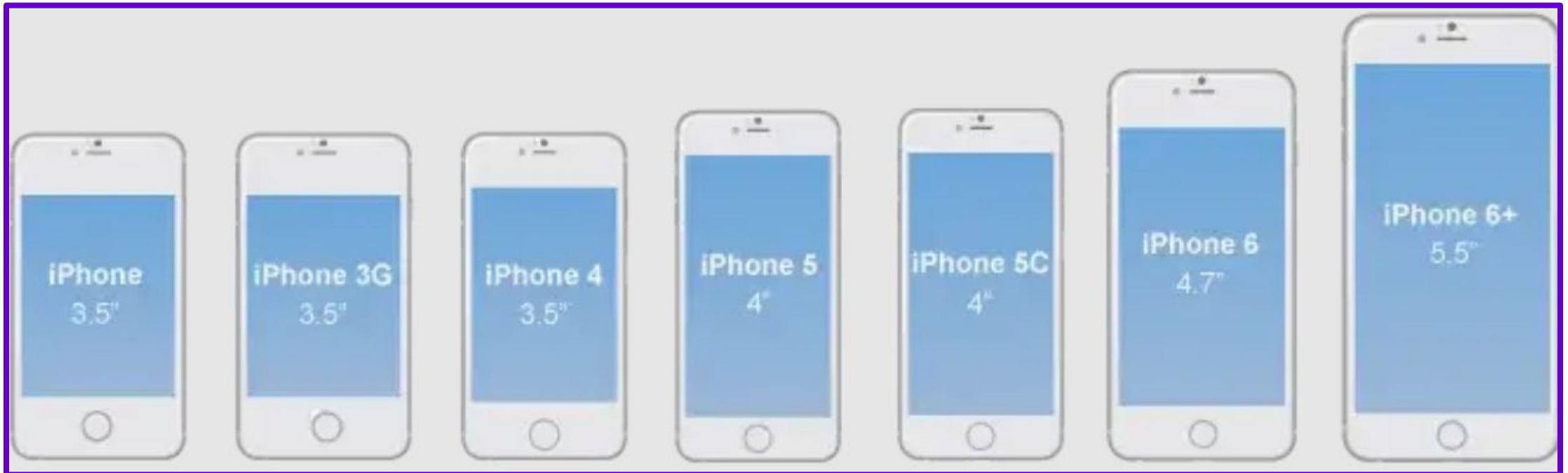
Layout and Grids

3

Design Objects

Screen Size

The screen size is the length of the screen diagonally or from one corner to the other.



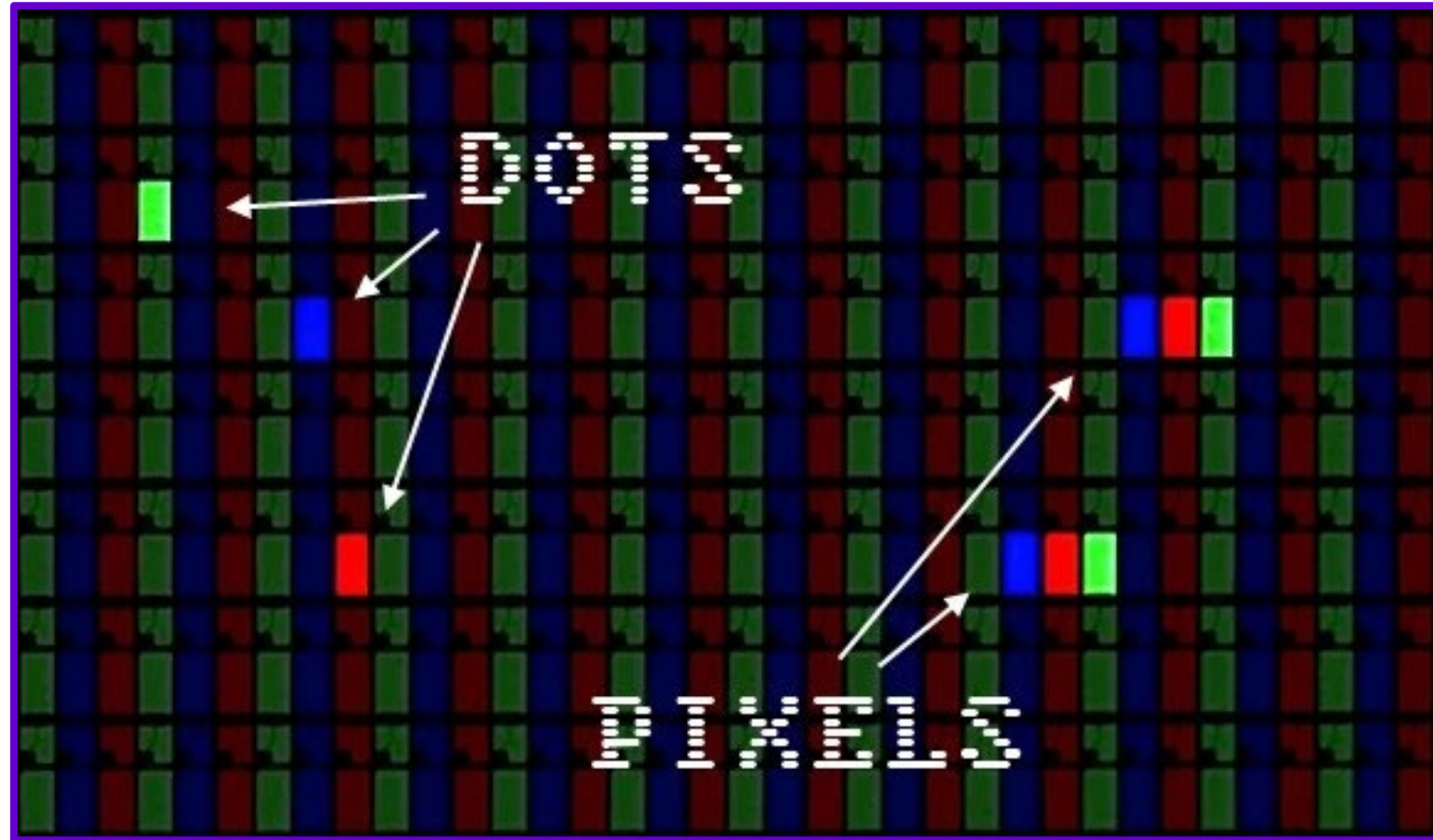
Screen Resolution

Resolution is the number of pixels going across the screen multiplied by the number of pixels going down.

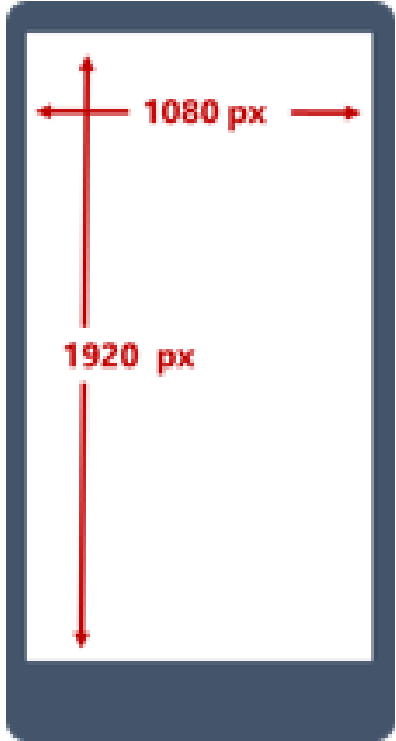
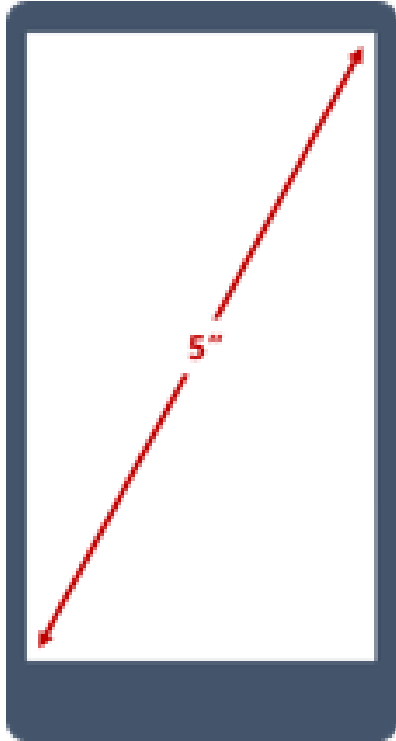
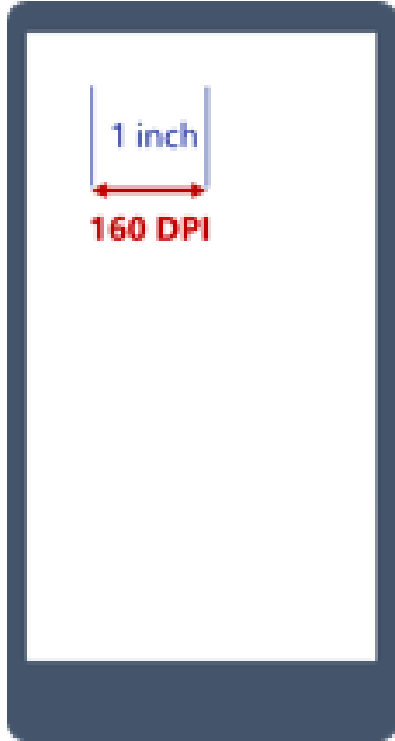


Pixels

A pixel is a single group of colored dots (normally red, green and blue) on a screen.



DPI, Size and Resolution

Screen Resolution	Size	Density (DPI)
The number of physical pixels in each direction	The diagonal size of the screen	The number of physical pixels/inch
		

Screens Evolutions

A typical late 90's CRT display had about 1/6 the resolution of the current iPhone that's much smaller in size.



Density Independent Pixels(DP) vs Point

Problem: how to design UI that Independent from devices resolutions?

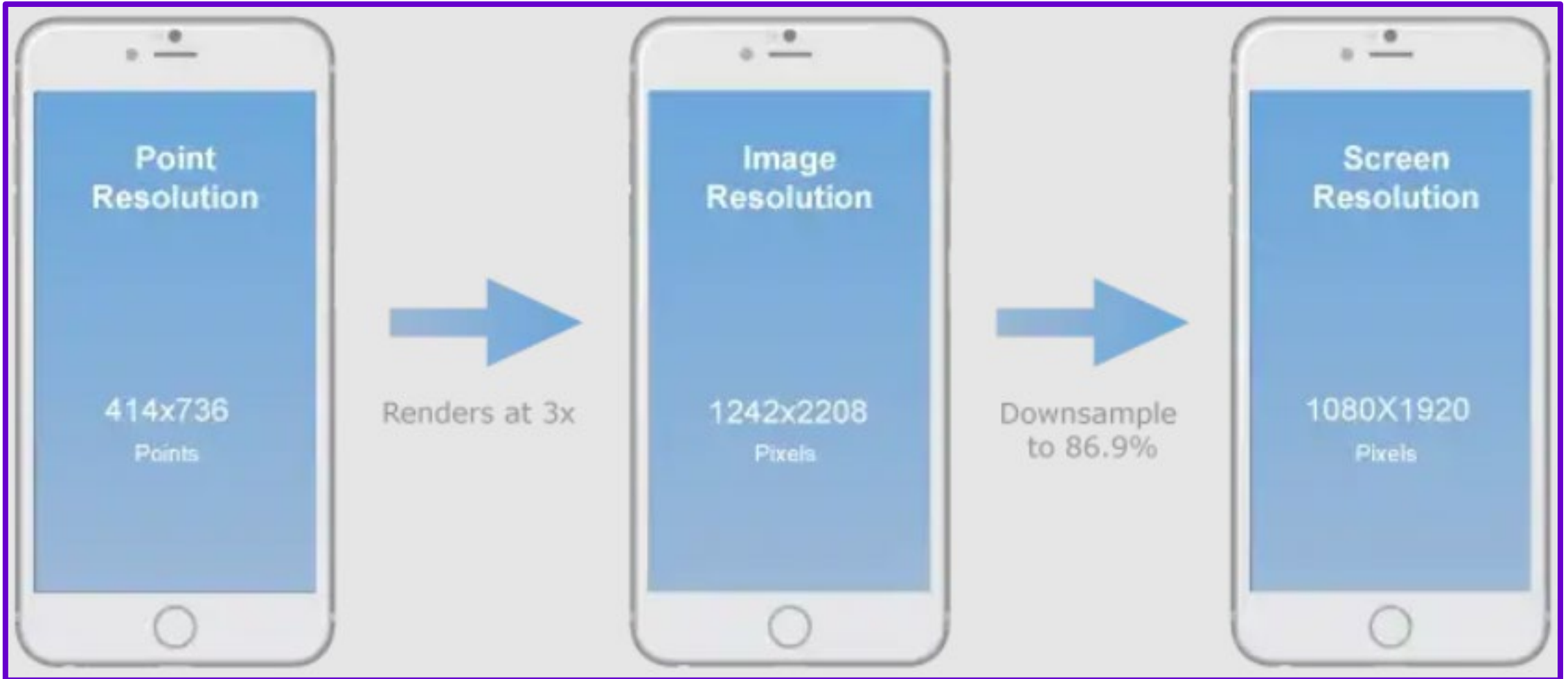


iOS Pixels = (Points * DPI) / 163



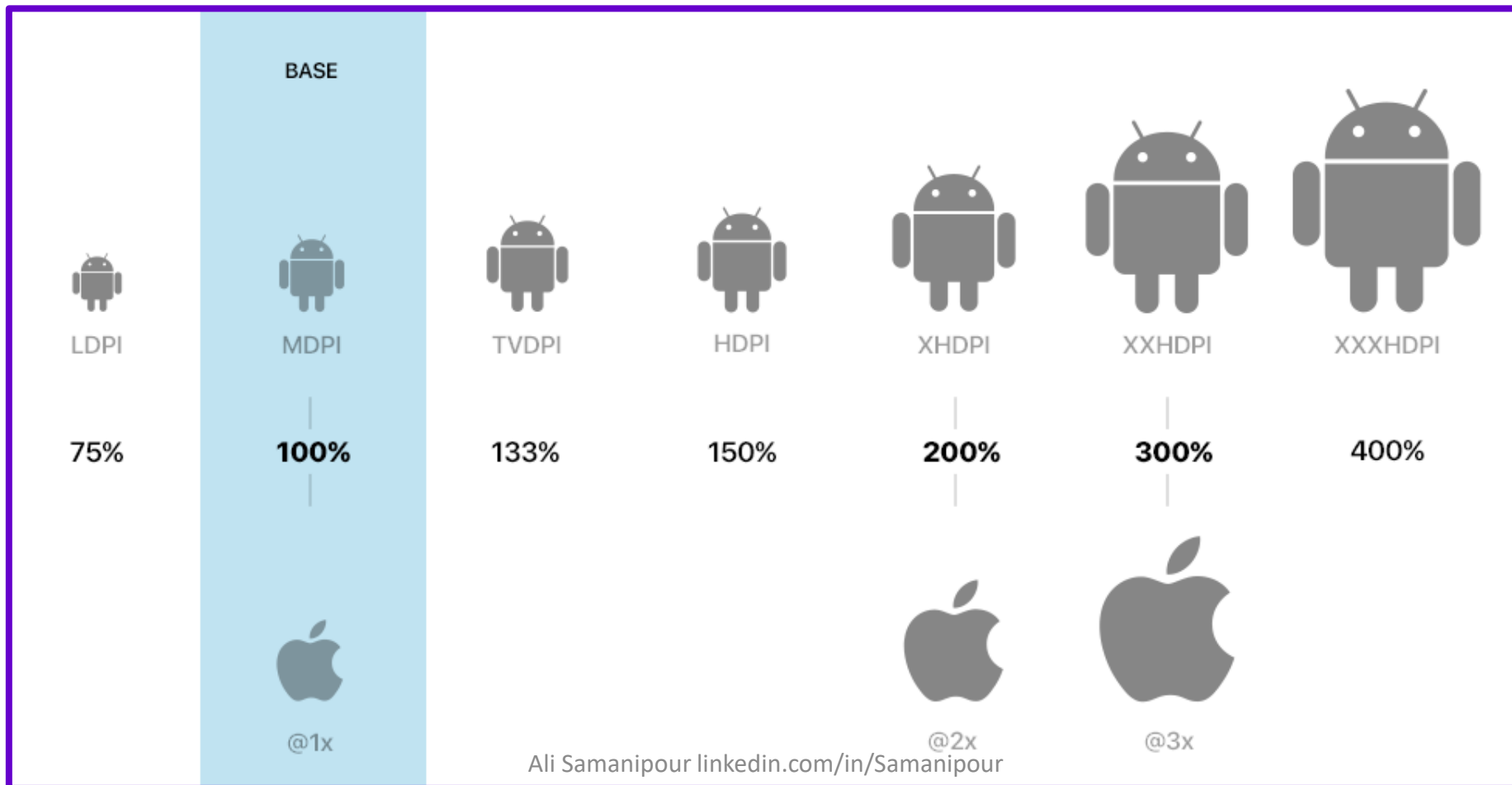
Android Pixels = (DP * DPI) / 160

Point, Image and Screen Resolution



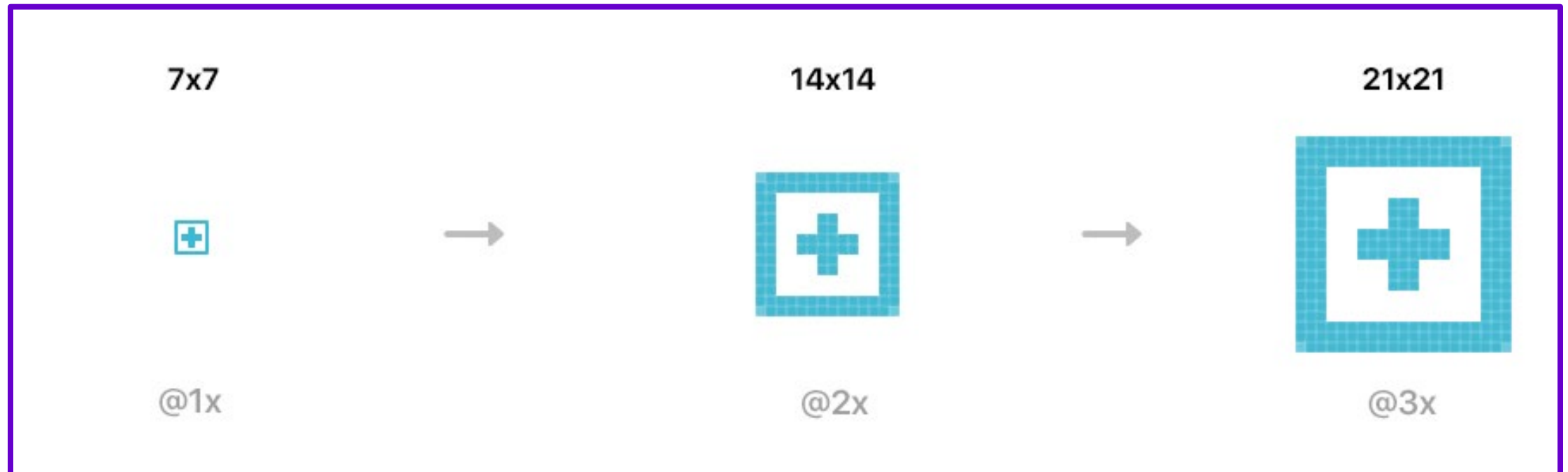
Achieve Resolution Independence

Always design for the base screen resolution or 1x.

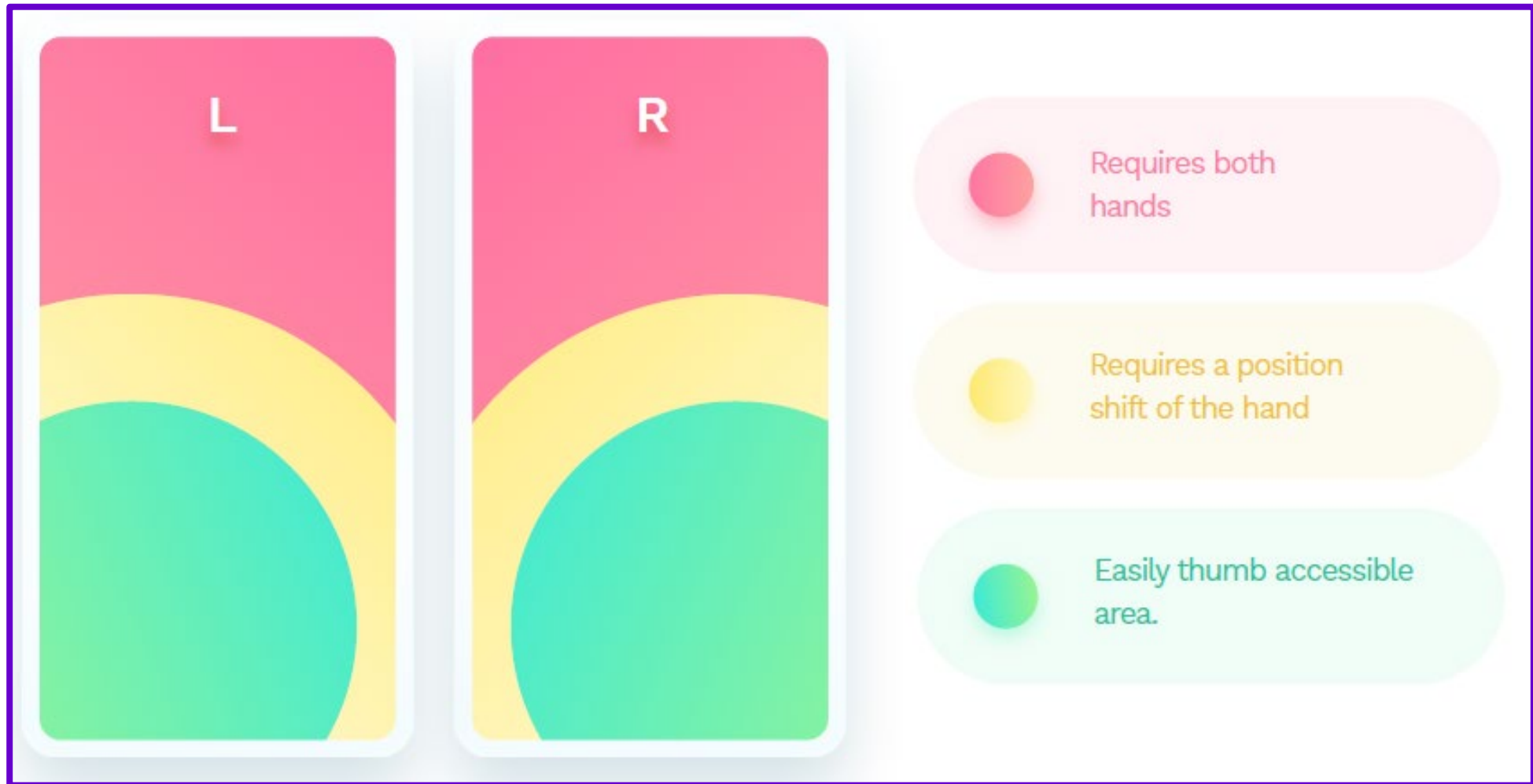


Benefits of 1x Design

improve scalability and ensure that you create “future proof” designs that will work in any resolution

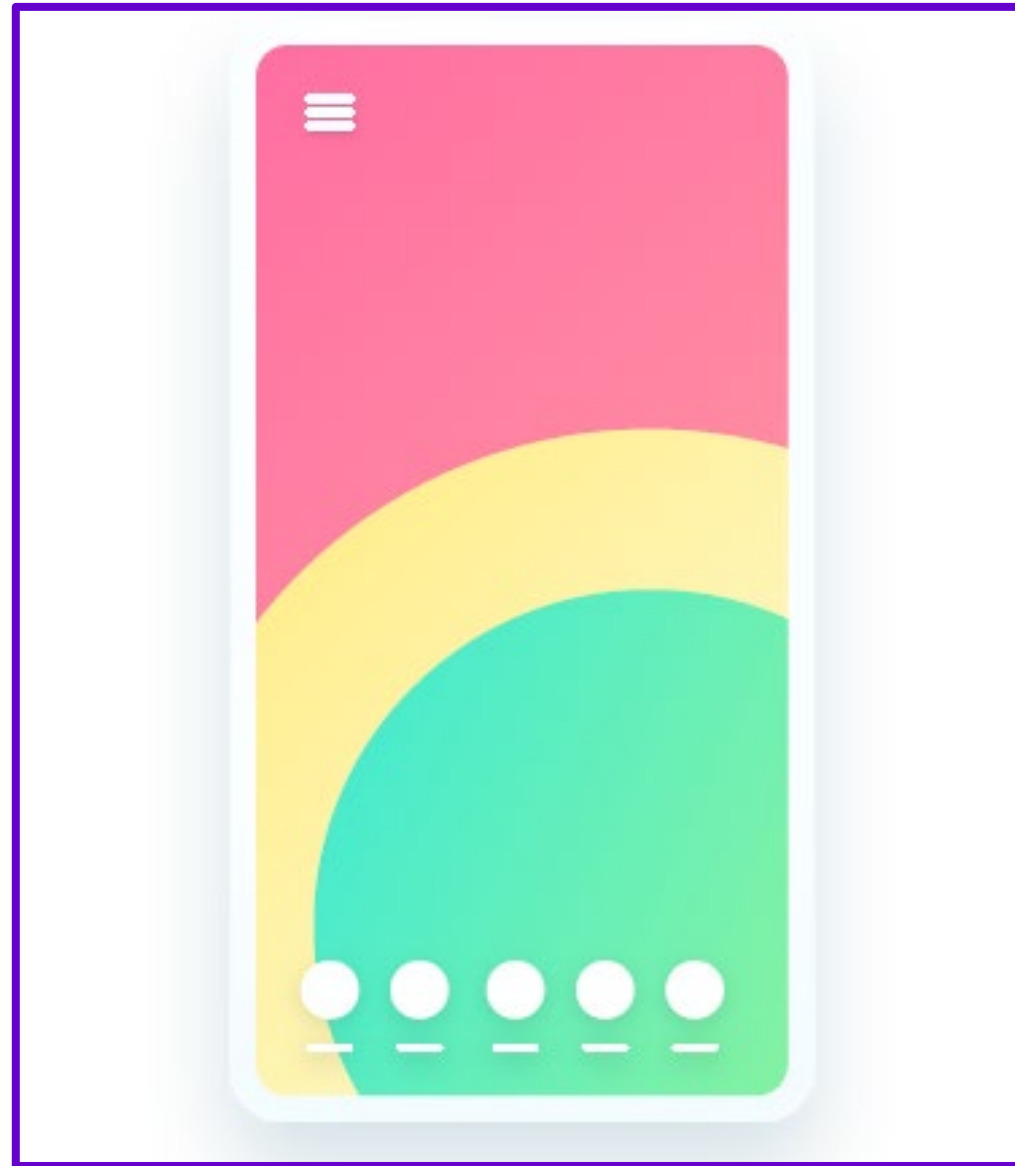


RANGE AND REACH



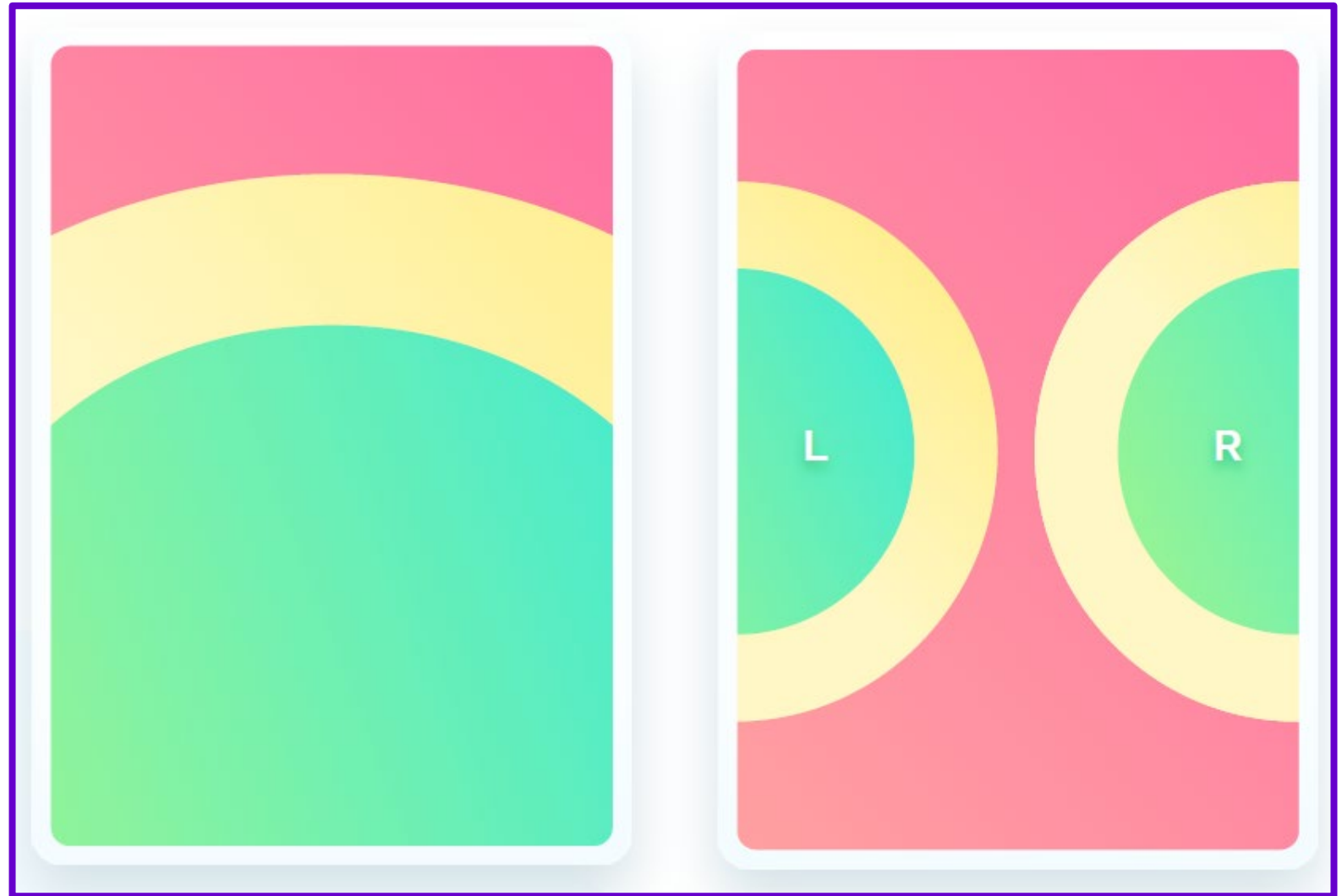
RANGE AND REACH

Reach help
determine how
easy it will be to
navigate the
product



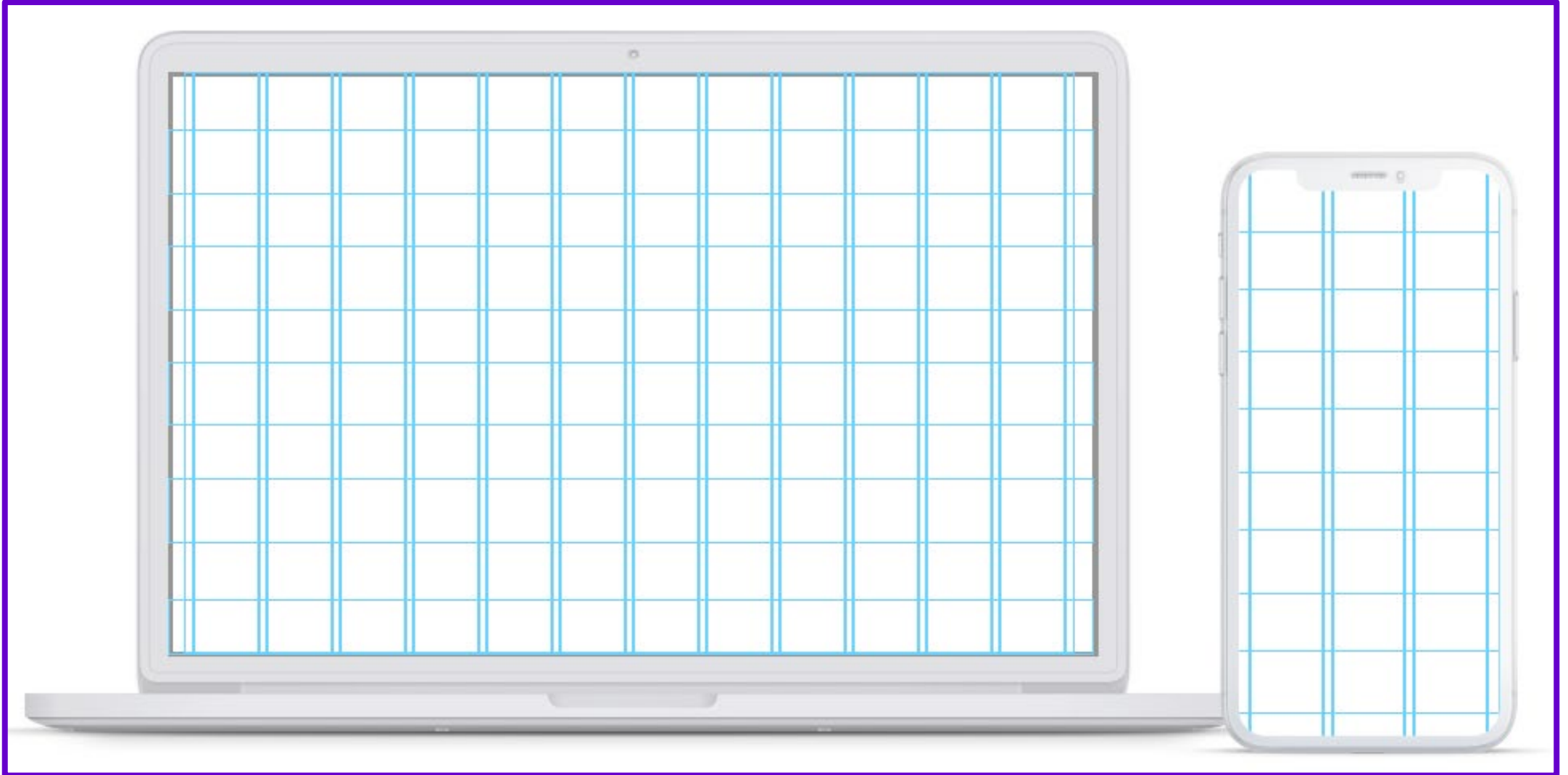
TABLET REACH

The most common way is one hand holding the device, while the other one is actively interacting with the UI



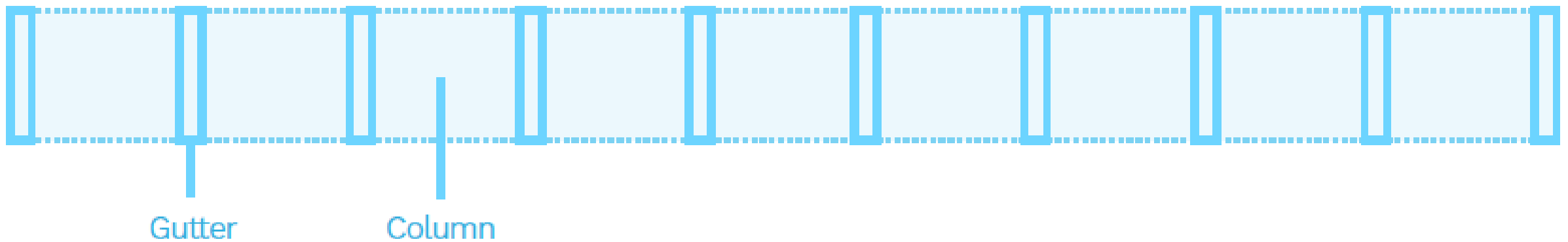
- 1 Screens
- 2 Layout and Grids
- 3 Design Objects

Layout & Grids



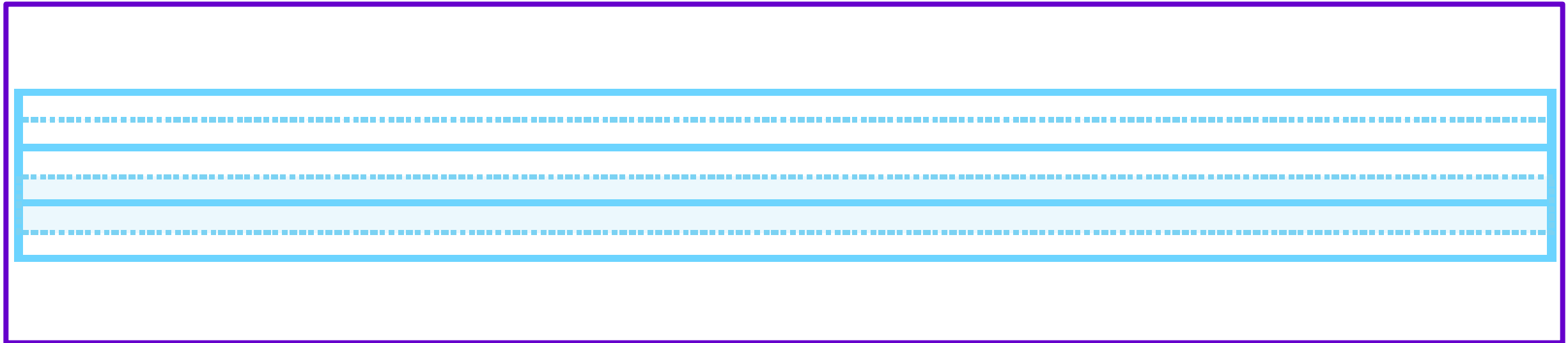
TYPES OF GRIDS (HORIZONTAL)

A horizontal grid is composed of vertical columns and margins between them, called gutters. Both of them can have either pre-set or flexible widths.



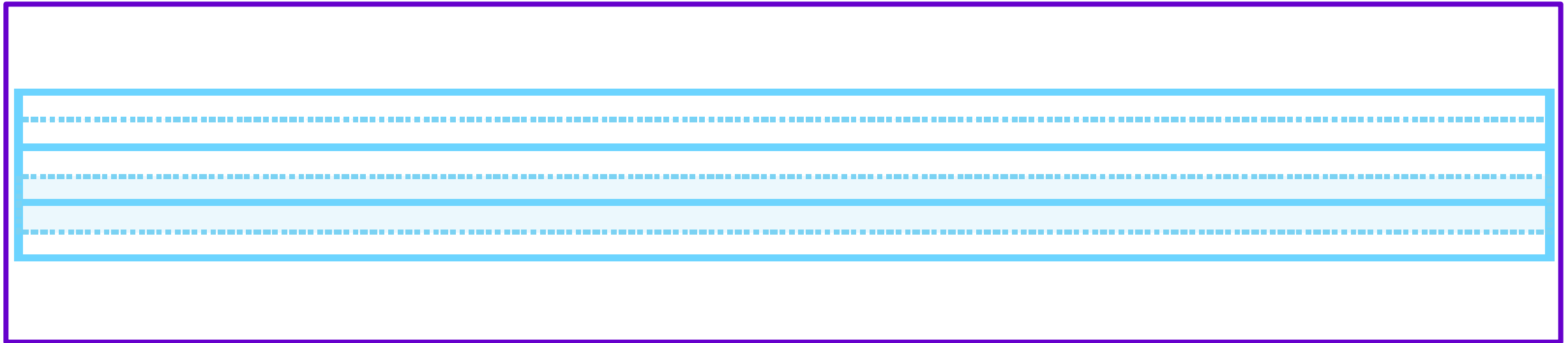
TYPES OF GRIDS (VERTICAL)

The vertical grid is less prevalent. It can help set the heights of elements, sections, and vertical whitespace.



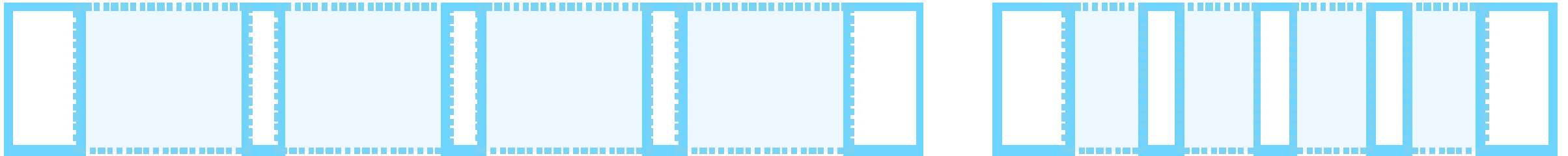
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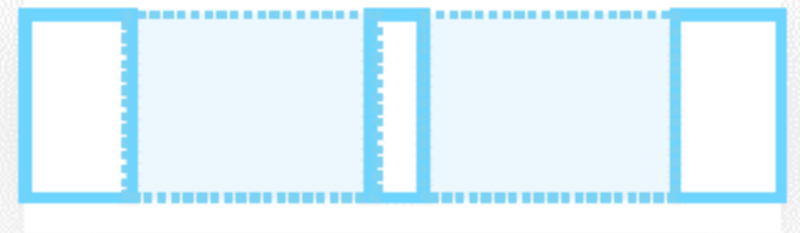
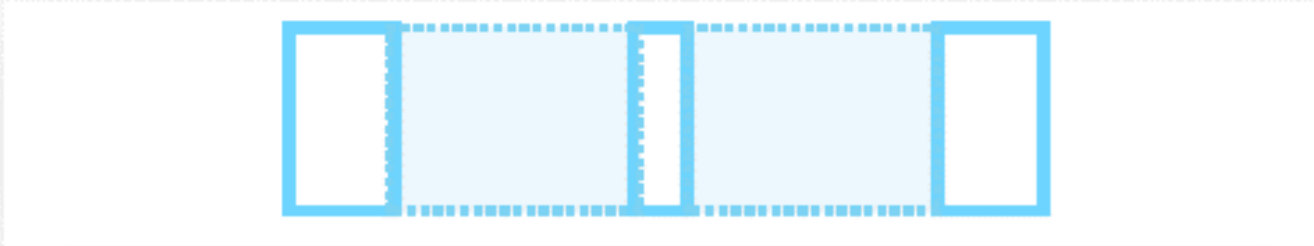
FLUID GRID

The fluid grid assumes an outer margin and gutter widths and adjusts the column widths to fit the screen. That makes the columns have varied width, and the gutters are what's keeping the alignment together



FIXED GRID

The fixed grid works with a set value for both the columns and gutter widths. If our screen is wider than the grid, we end up with blank space on the sides.



BASE VALUE (10 POINT GRID)

Building a grid should always start with choosing its base value. It's the smallest number we use to set all the other values. In general, all our grid values should be divisible by that base number.

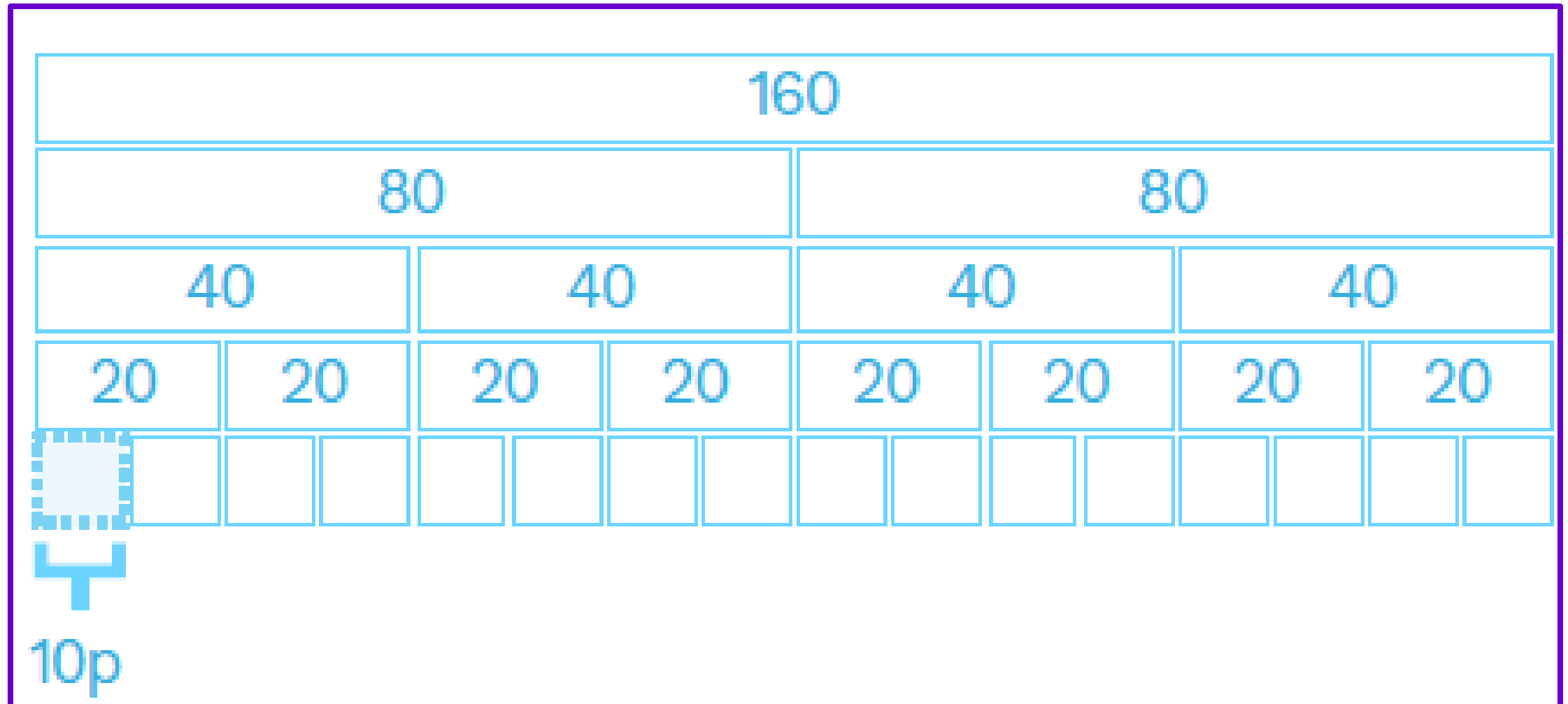


BASE VALUE (10 POINT GRID)

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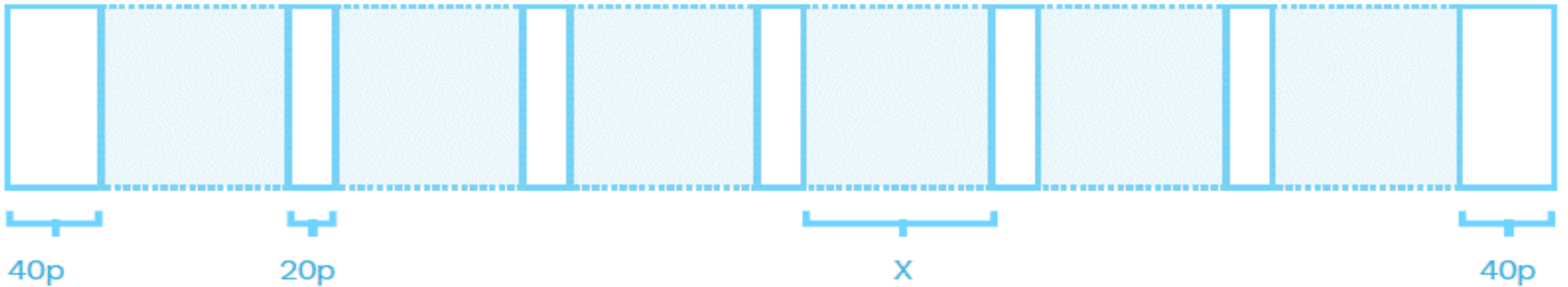


10 POINT GRID SYSTEM



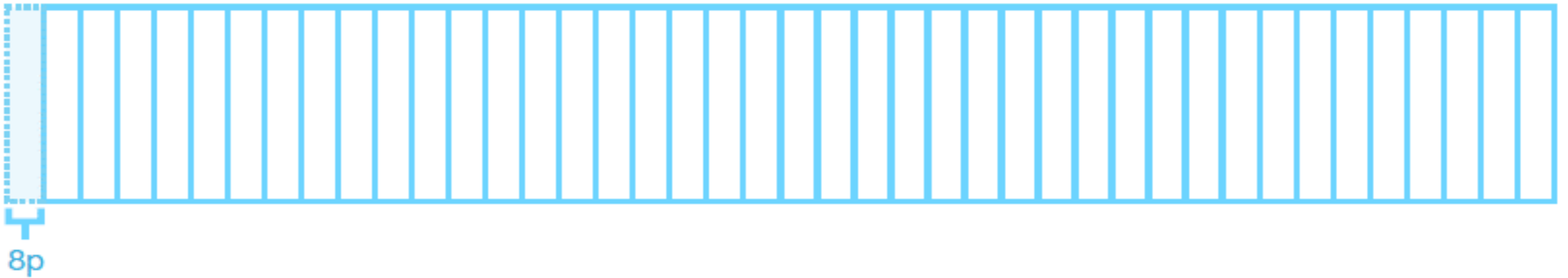
10 POINT FLUID GRID

As ten itself is probably too small of a number to work well as our gutter (for readability), we go with double that for gutter and quadruple for outer margins.

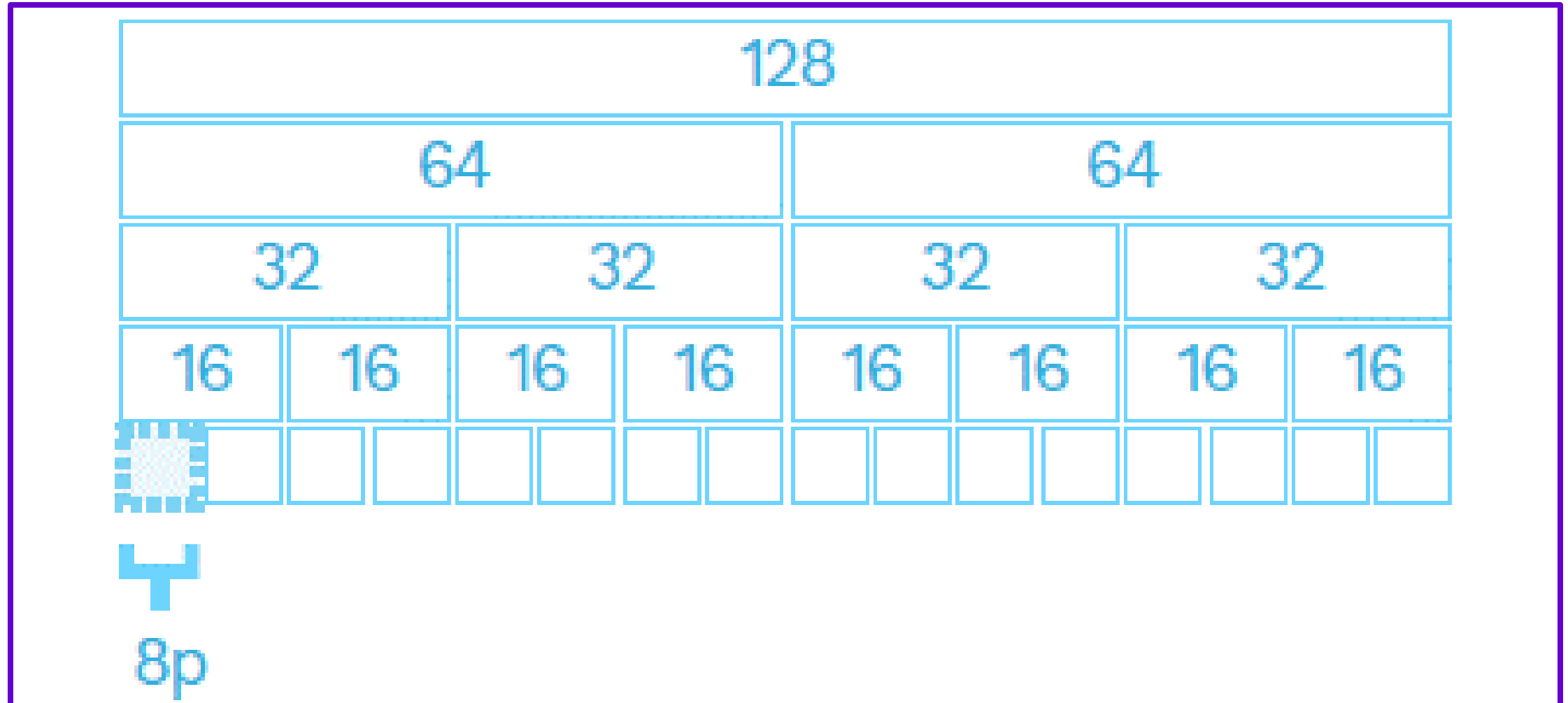


BASE VALUE (8 POINT GRID)

The 8-point grid is currently the most popular grid-type in modern UI design. It requires a bit of extra effort, but it's well worth trying it.

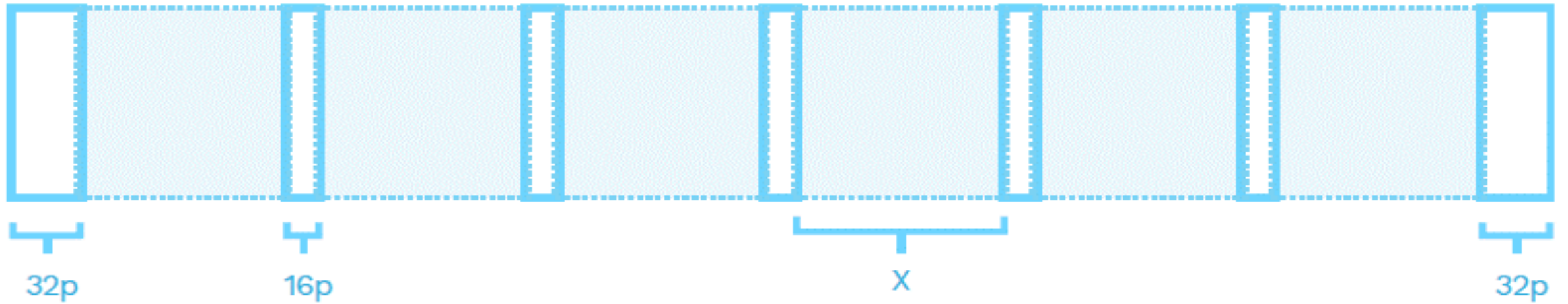


8 POINT GRID SYSTEM



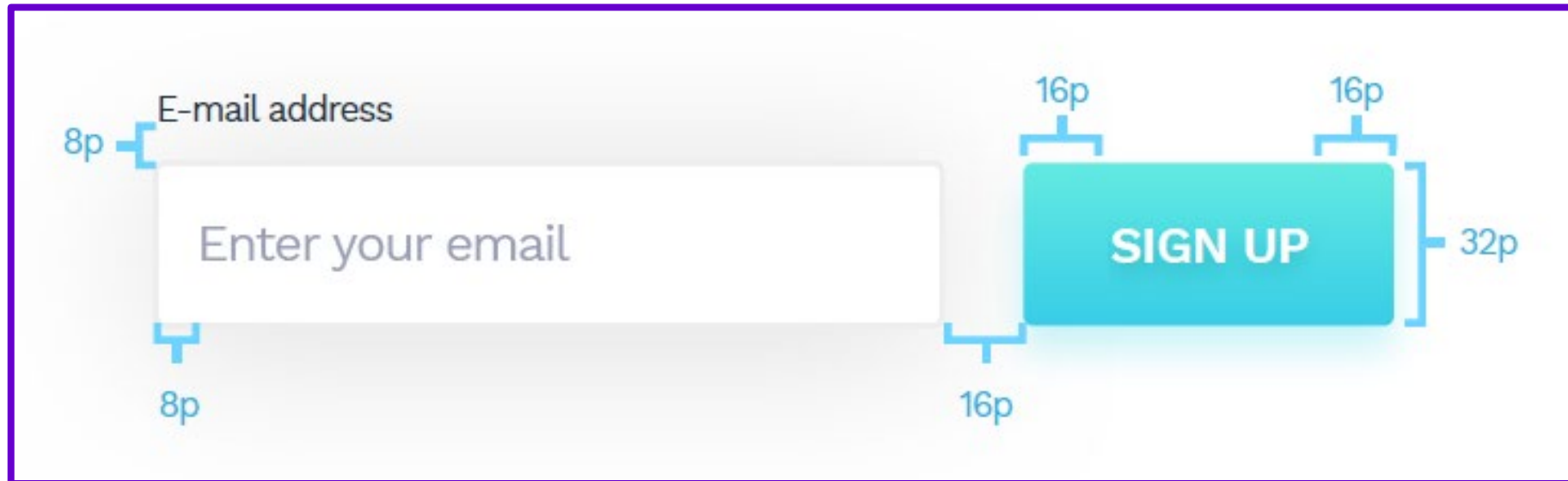
8 POINT FLUID GRID

Creating an 8-point fluid grid should start with multiplying the base number by two. Eight in itself is too small to be the right choice for separating objects



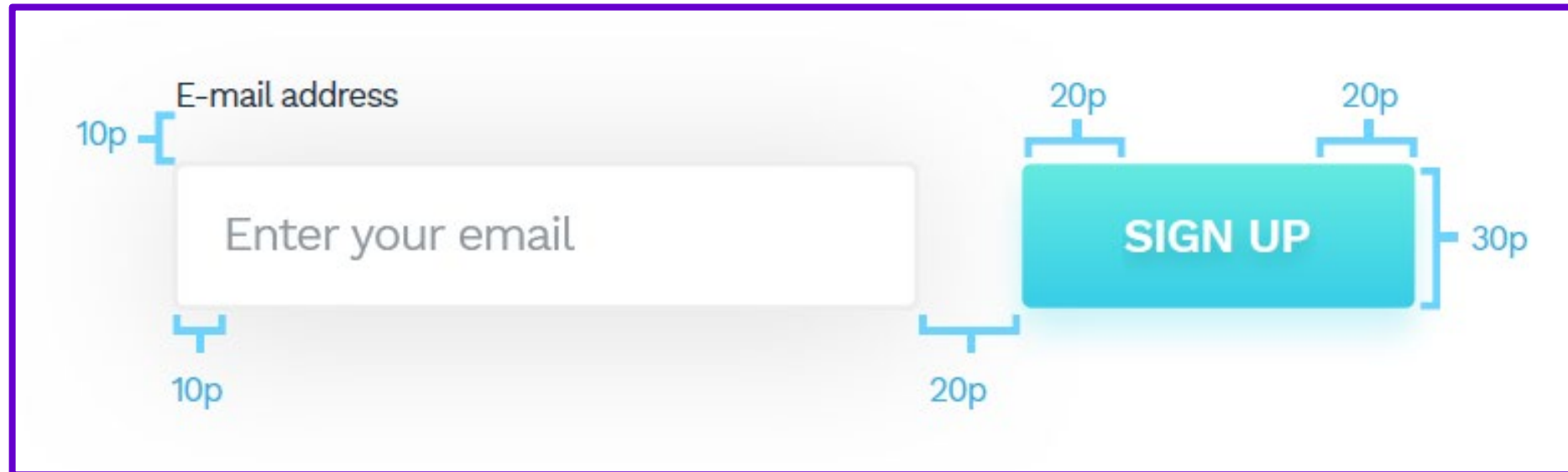
WHY 8 IS BETTER THAN 10?

One of the primary reasons is the fact that if we use the multiples of 8, it allows us to have more freedom with setting both external and internal margins in an object

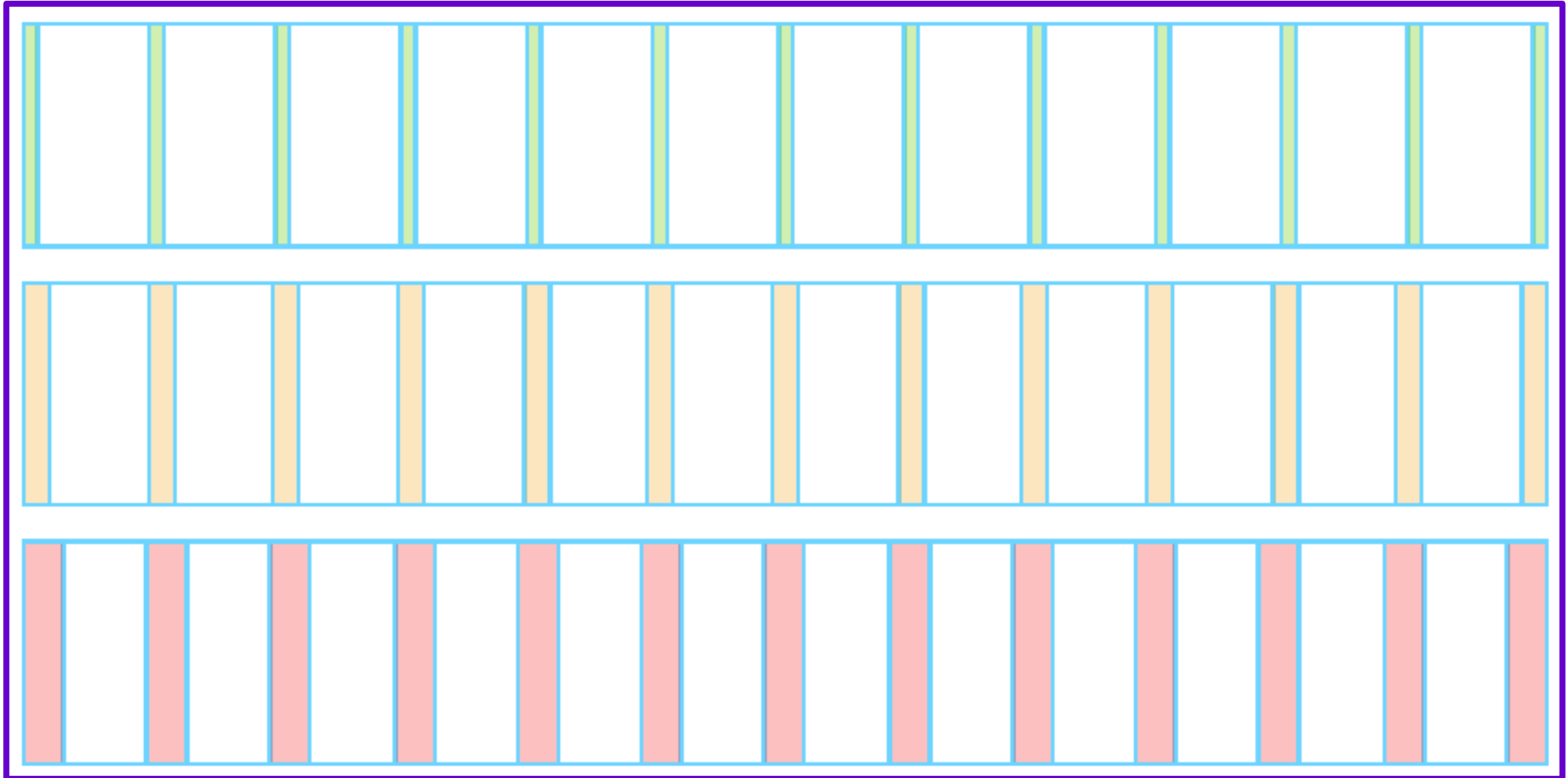


WHY 8 IS BETTER THAN 10?

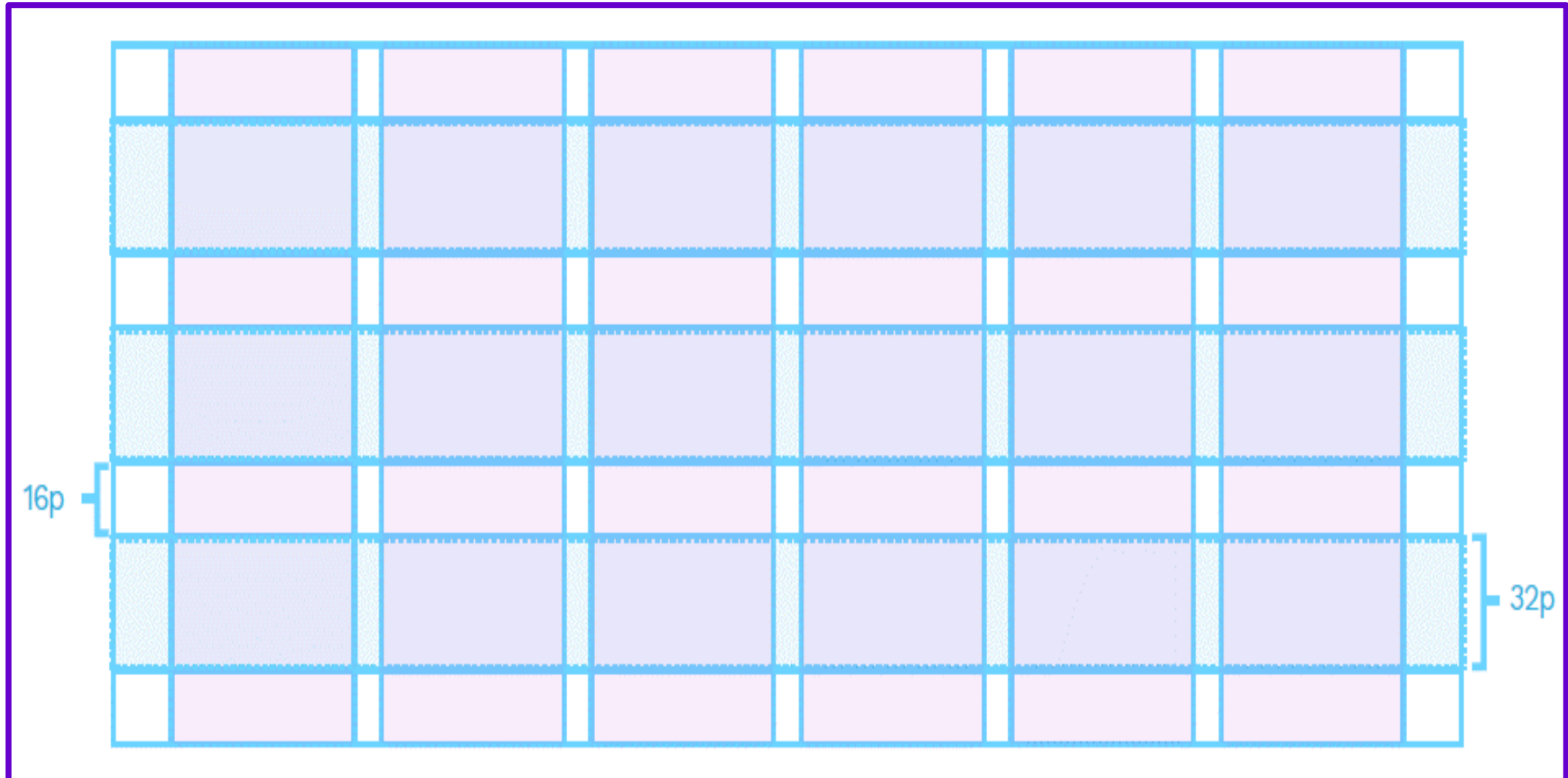
When our minimum nudge value equals ten, it forces us to have much larger margins (of 20).



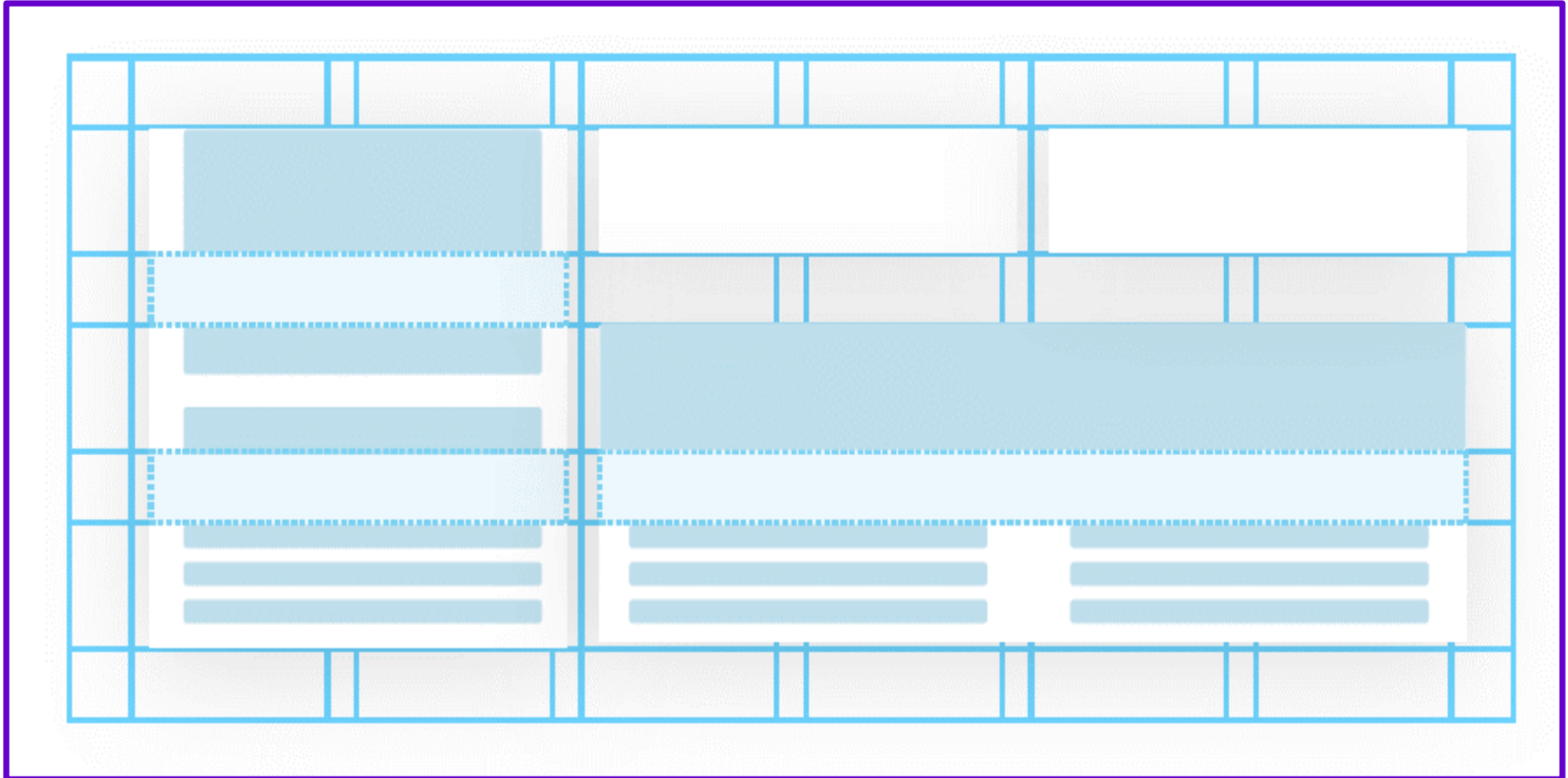
MULTIPLE GRIDS



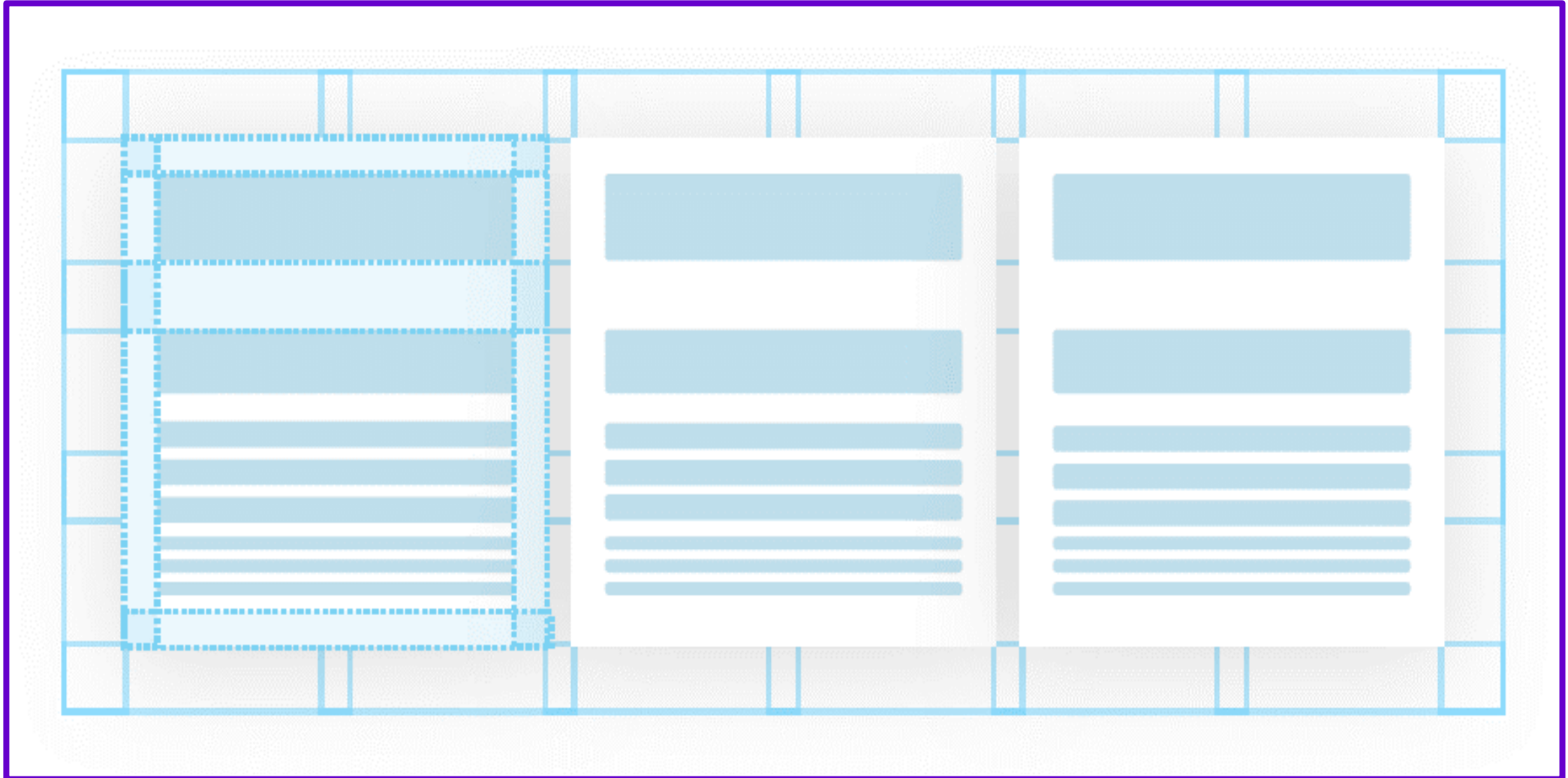
VERTICAL RHYTHM



ALIGNING OBJECTS TO HORIZONTAL AND VERTICAL GRIDS

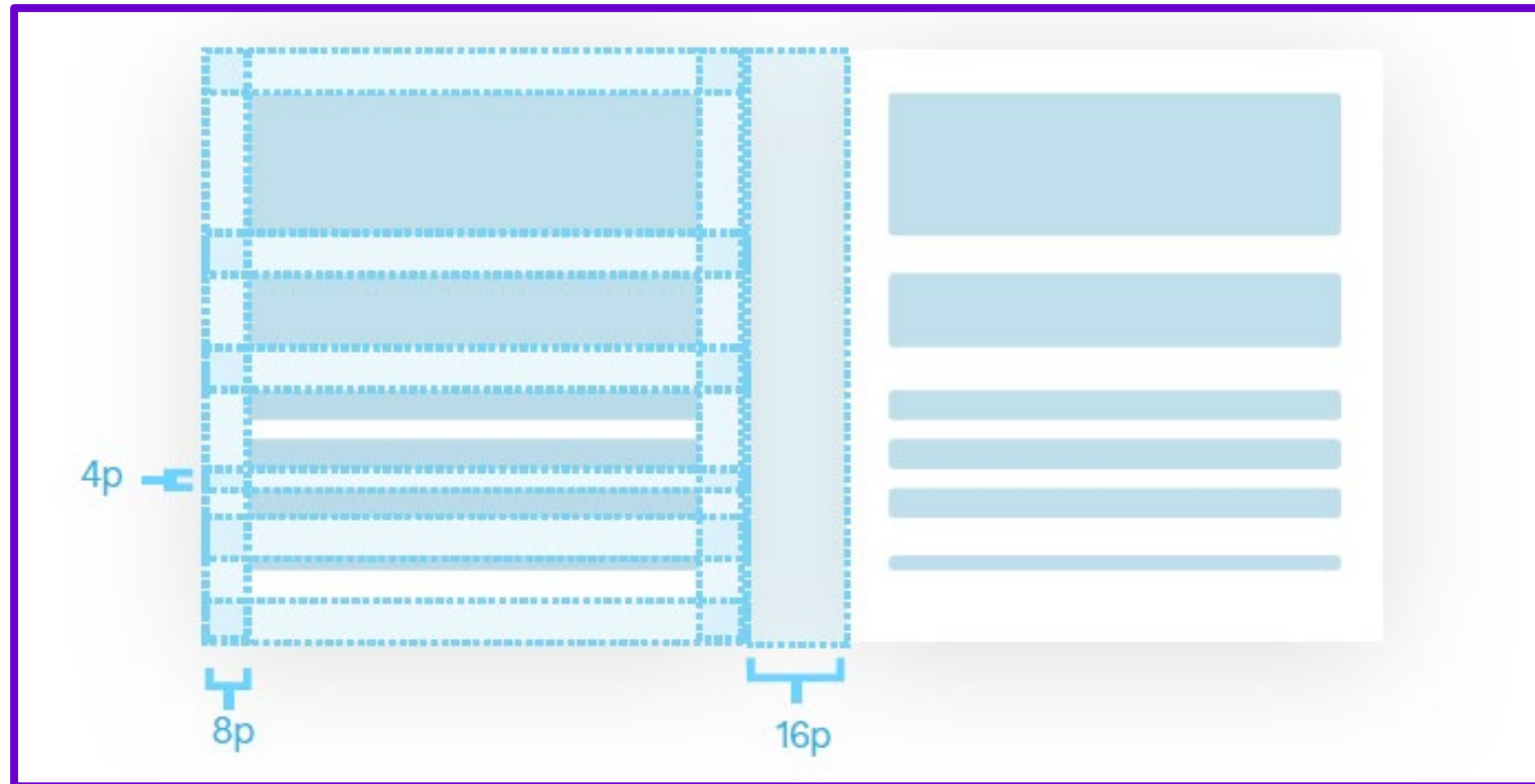


COMPONENT

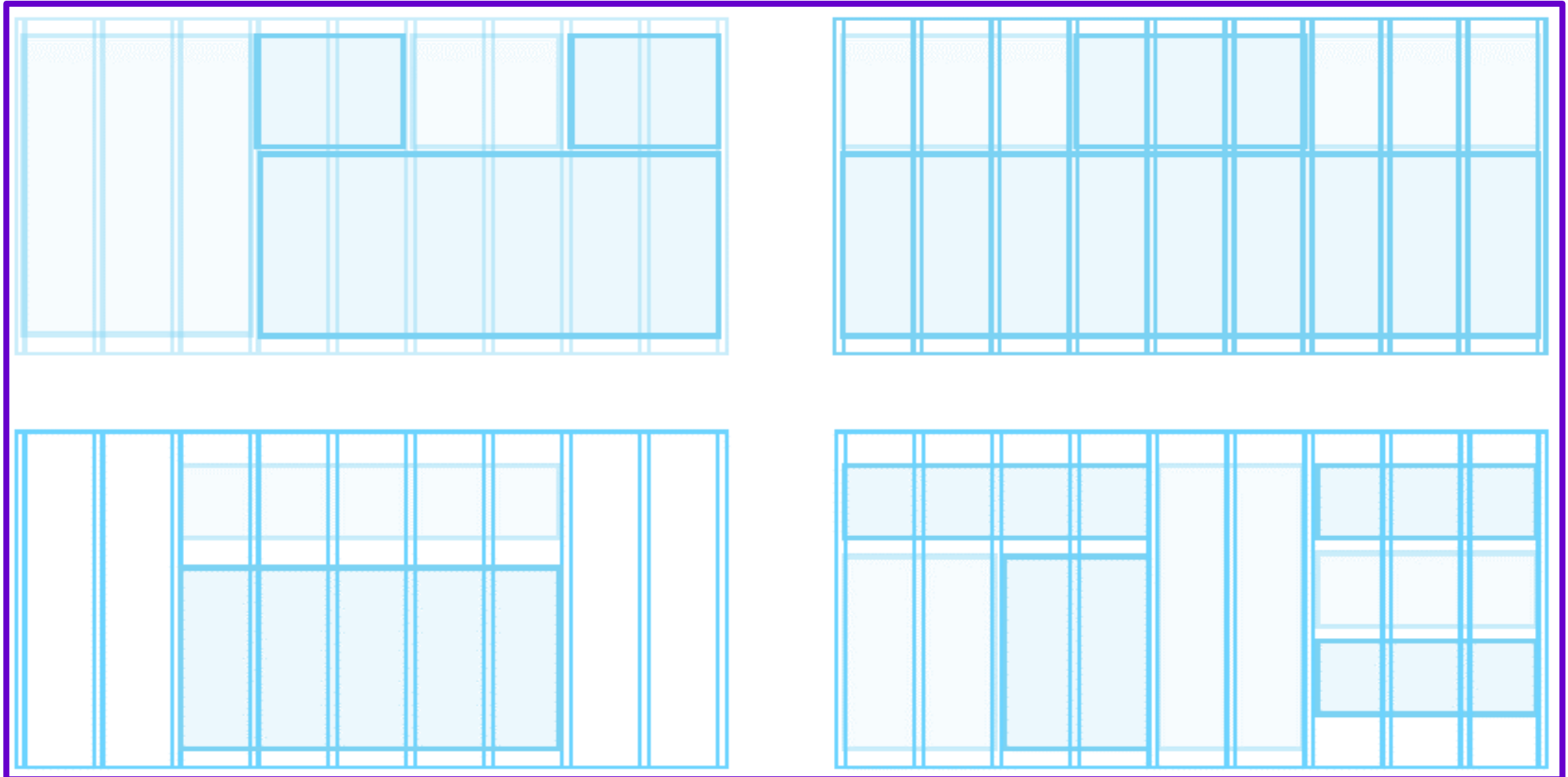


SOFT GRID

A soft grid is a rule of aligning objects using the base number and its multiplies

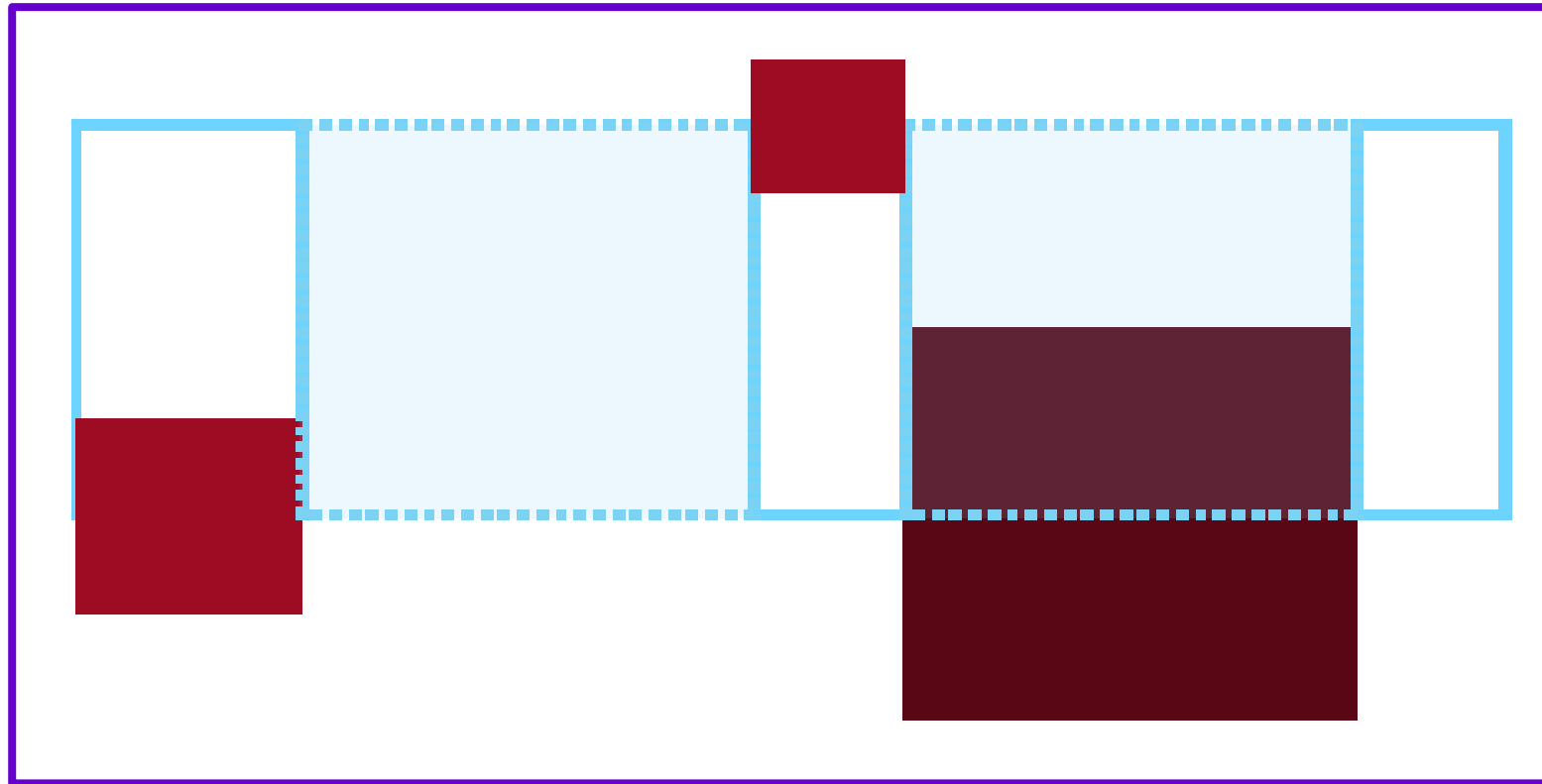


HOW TO CREATE A GRID?

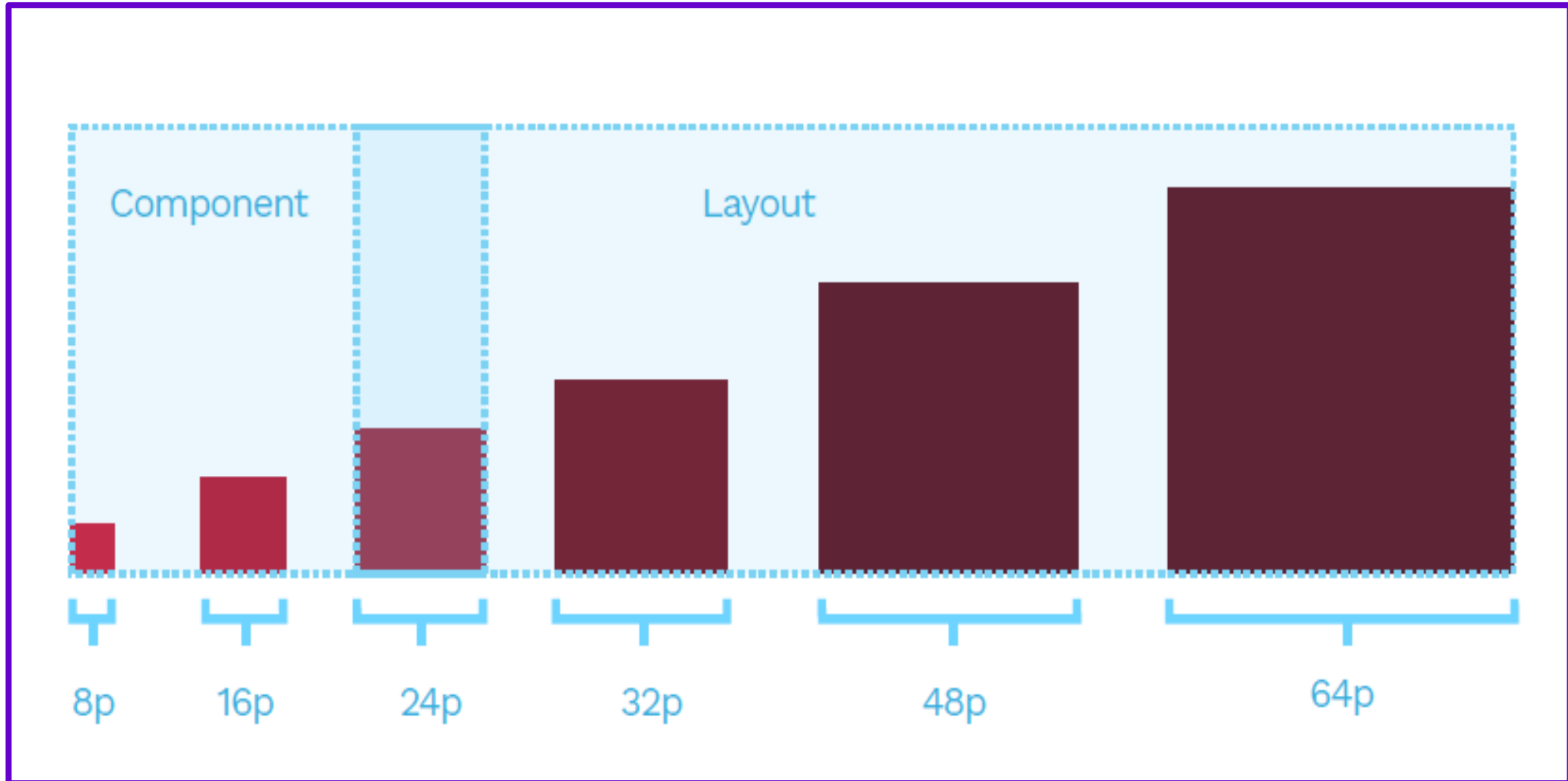


THE RED SQUARE METHOD

A Red Square is a full shape that you can place between the lines of any grid to check alignment.



THE RED SQUARE METHOD (Base Point)



THE RED SQUARE METHOD



Random Soda

\$5.00

Refreshing mint-flavored
soda with no added sugar.

+ ADD TO CART



Random Soda

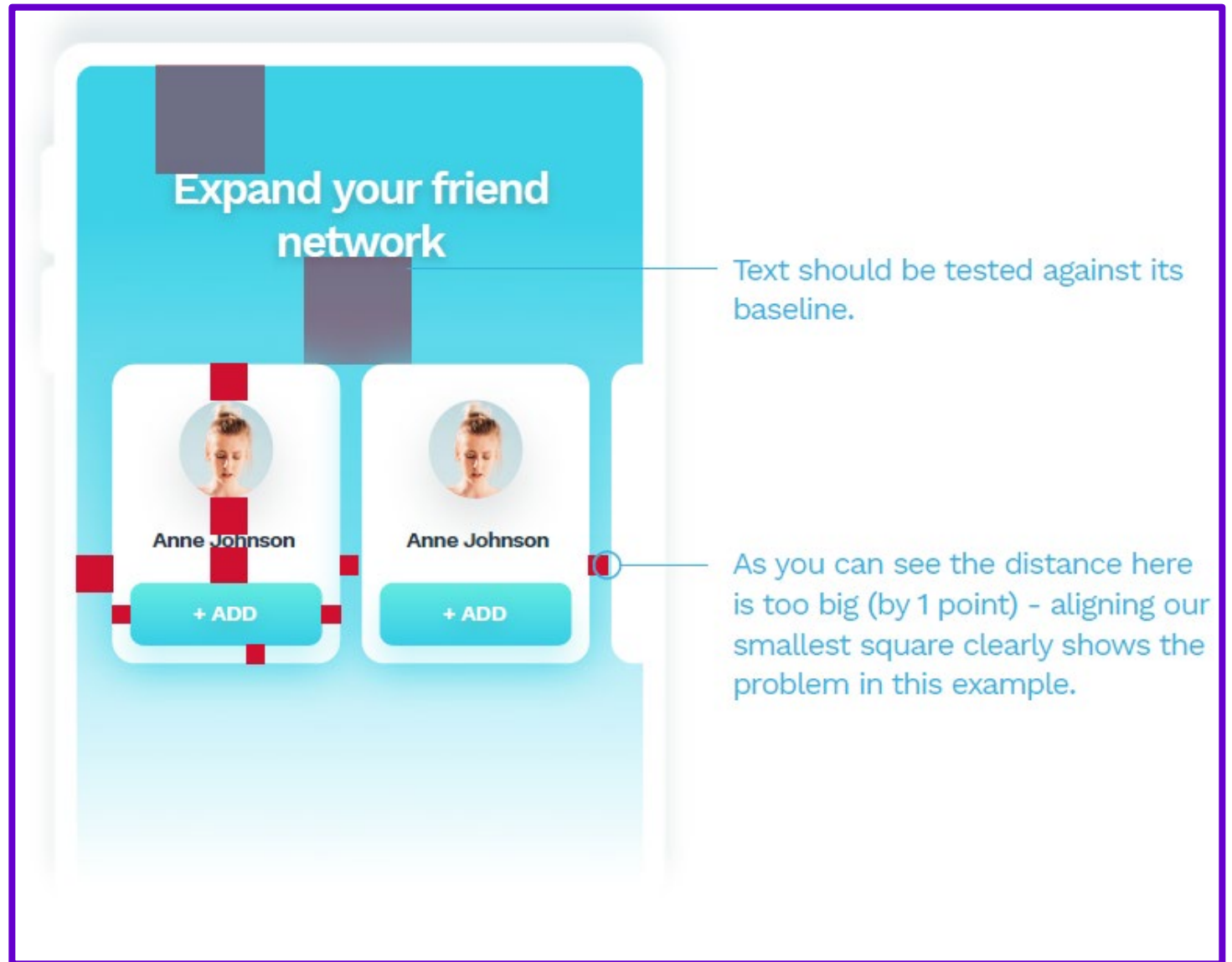
\$5.00

Refreshing mint-flavored
soda with no added sugar.

+ ADD TO CART

Auditing an existing design

looking for is
deviations from the
rules



TEXT ALIGNMENT

The Matrix

Release date: 1999

Computer programmer Thomas Anderson, known in the hacking scene by his alias "Neo", feels something is wrong with the world and is puzzled by repeated online encounters with the phrase "the Matrix." Trinity contacts him and tells him a man named Morpheus has the answers he seeks.

Categories

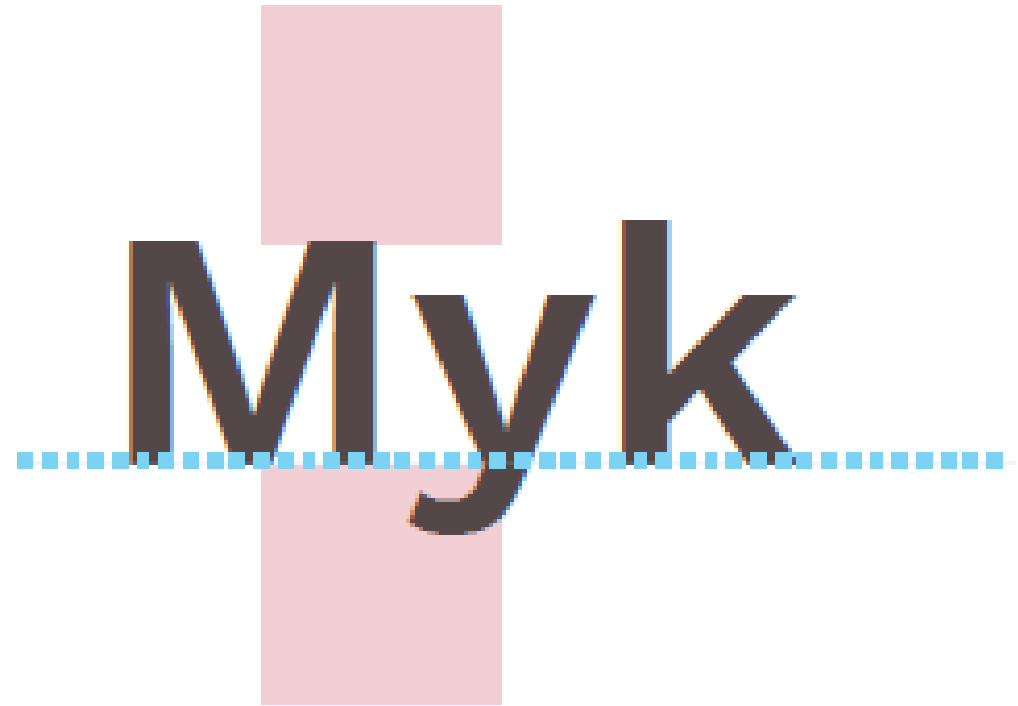
Action

Sci-fi

Cyberpunk

TEXT ALIGNMENT

We are aligning our squares to the CAP-height which also means they align to the baselines of our font.



Some Best Practices

Layouts should be simple. In web design projects start with either 12 or 16 columns for greater flexibility

Longer text (body copy) on a 12-column grid should not exceed eight columns

DEFINE THE GRID ASAP!

HOW WE SCAN CONTENT (F-Pattern)

We don't read
anymore.
We scan.

The most
popular
across
Europe and
The US



HOW WE SCAN CONTENT (Z-Pattern)

We don't read
anymore.
We scan.

Z-pattern
happens
when a large
photo or
video breaks
an F-pattern.



1

Screens

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Layout and Grids

3

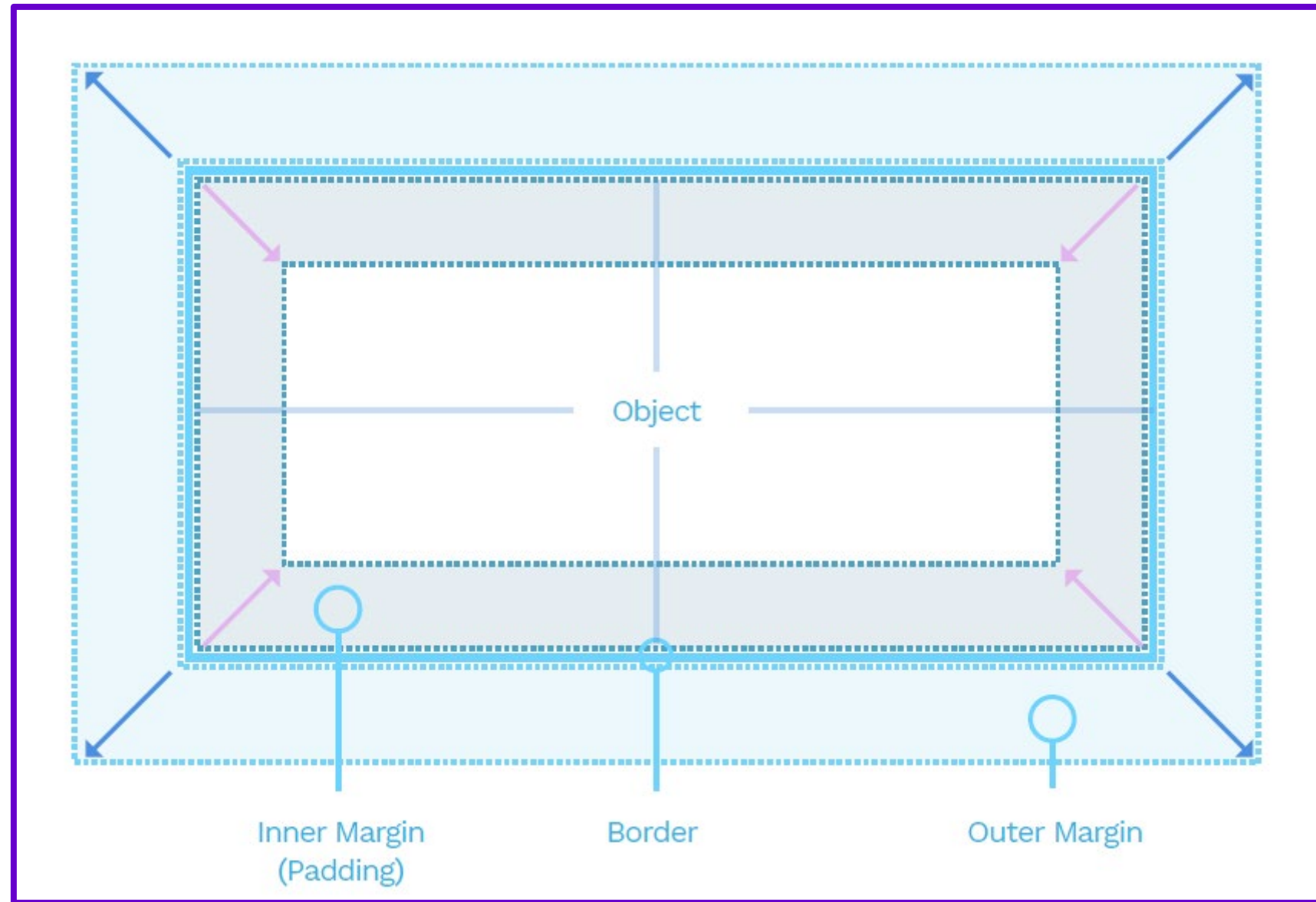
Design Objects

Objects

The general idea of UI design is about moving rectangles around in just the right way



THE BOX MODEL

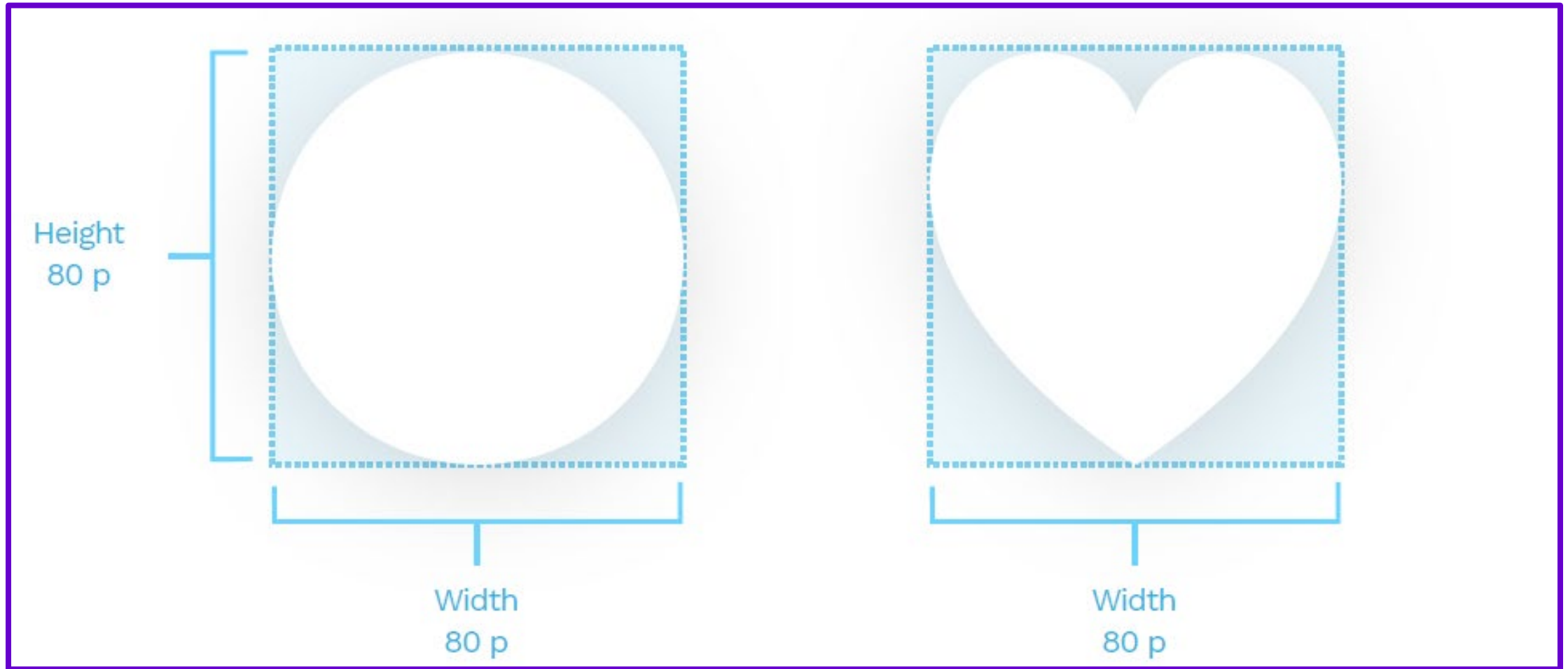


SIZE

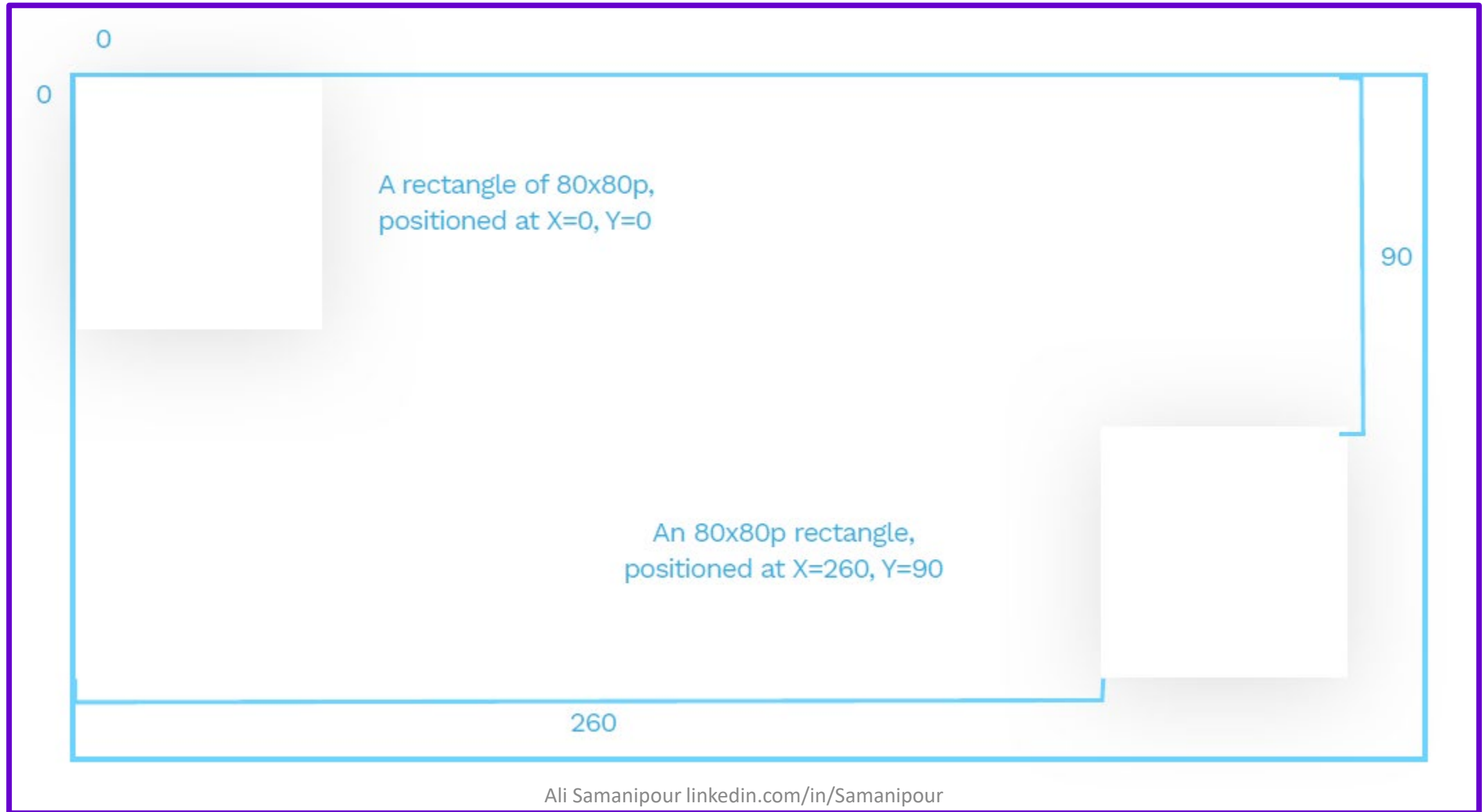
Height or H
80p

Width or W
120p

SIZE



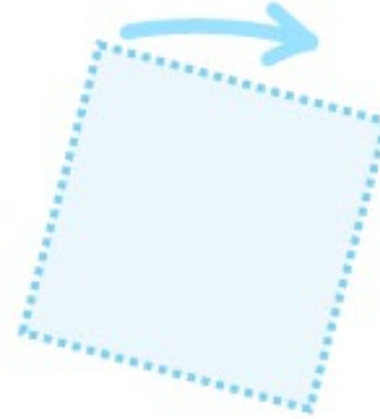
POSITION



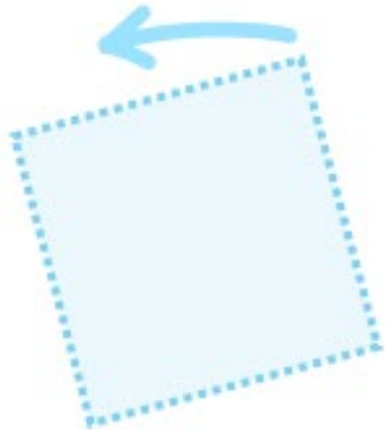
ANGLE



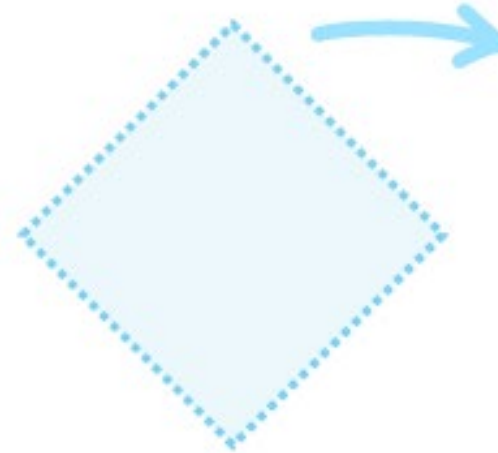
0° Rotation



15° Rotation



-15° Rotation
(or 345° Rotation)



45° Rotation

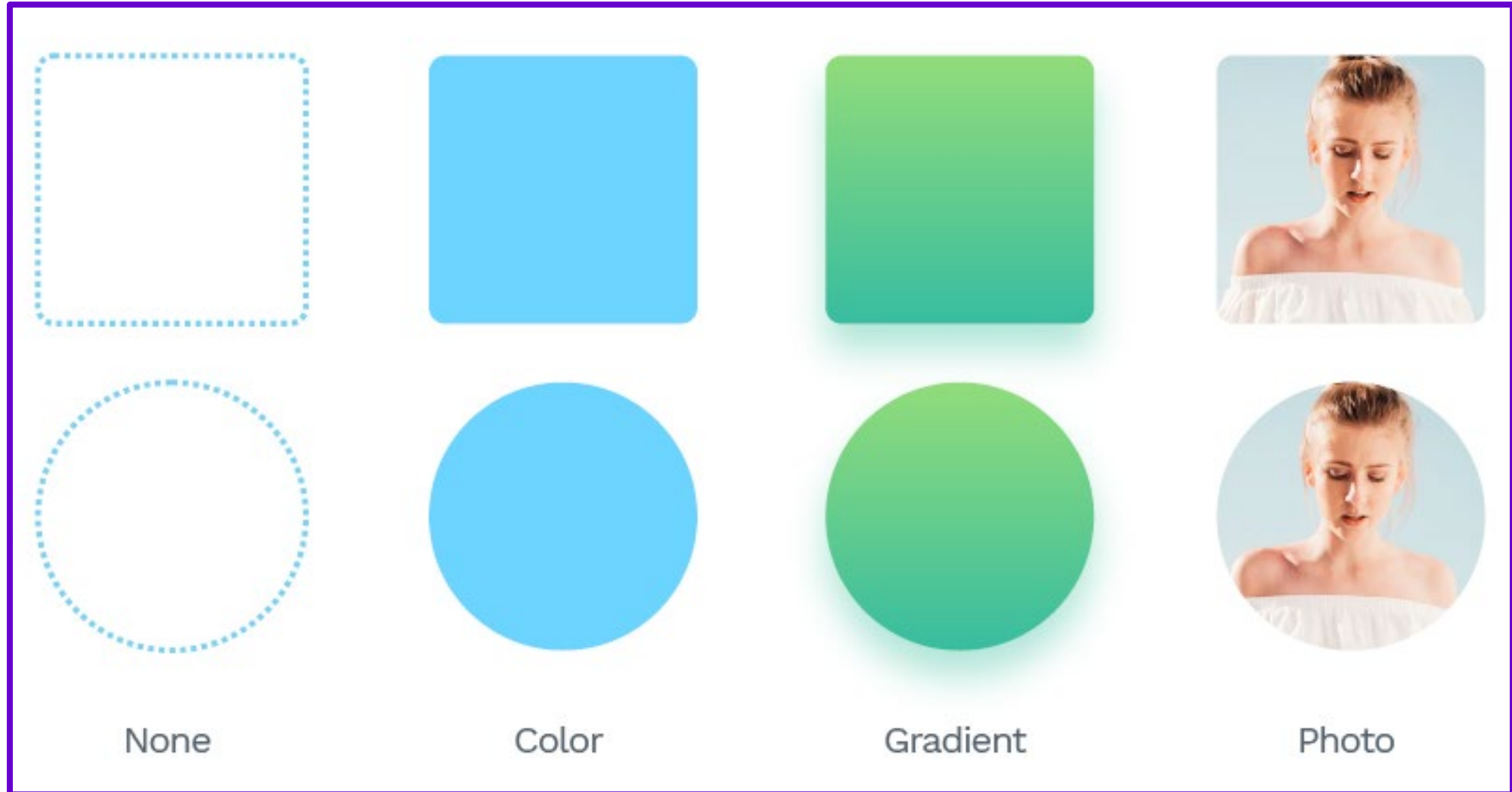
BORDER-RADIUS



Starting from the left a border-radius of : 0p , 2p , 6p , 100p



FILL



FILL...



None



Color

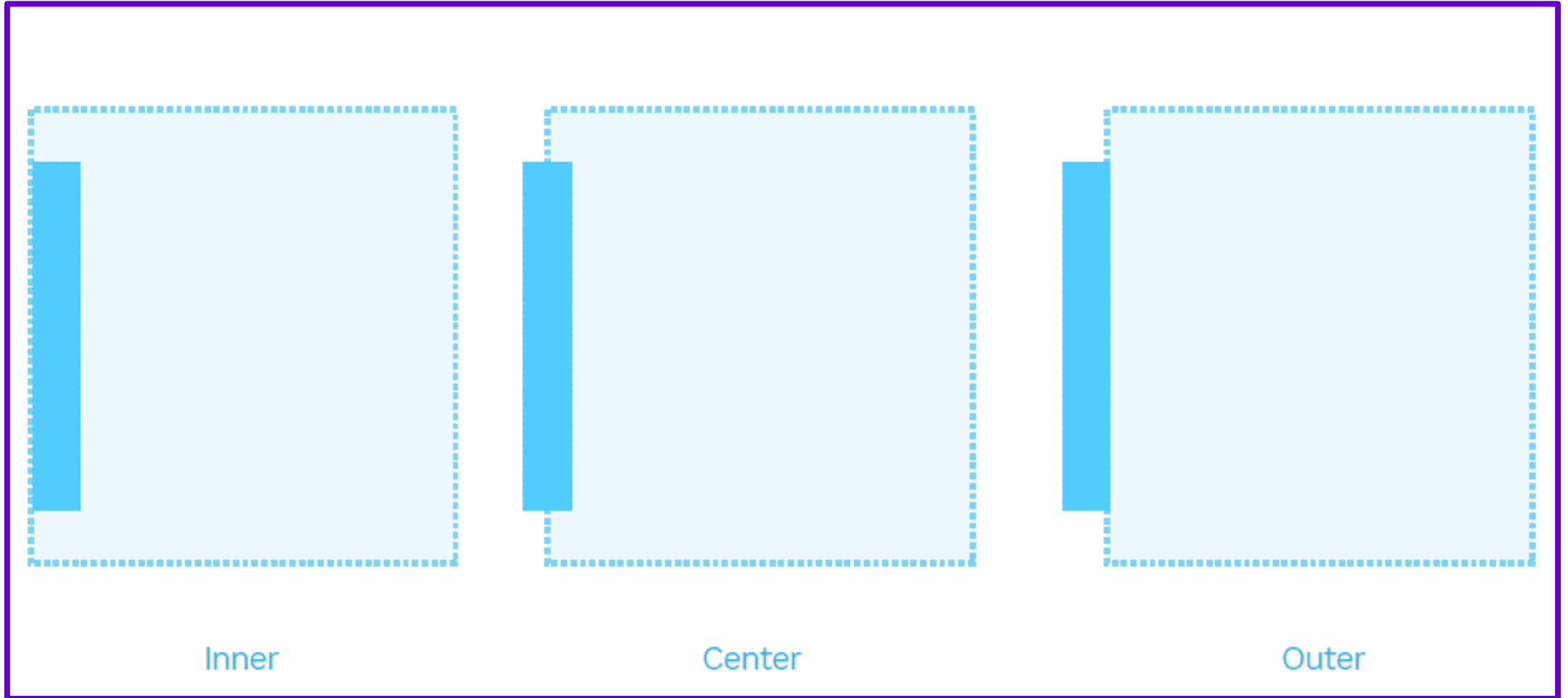


Gradient

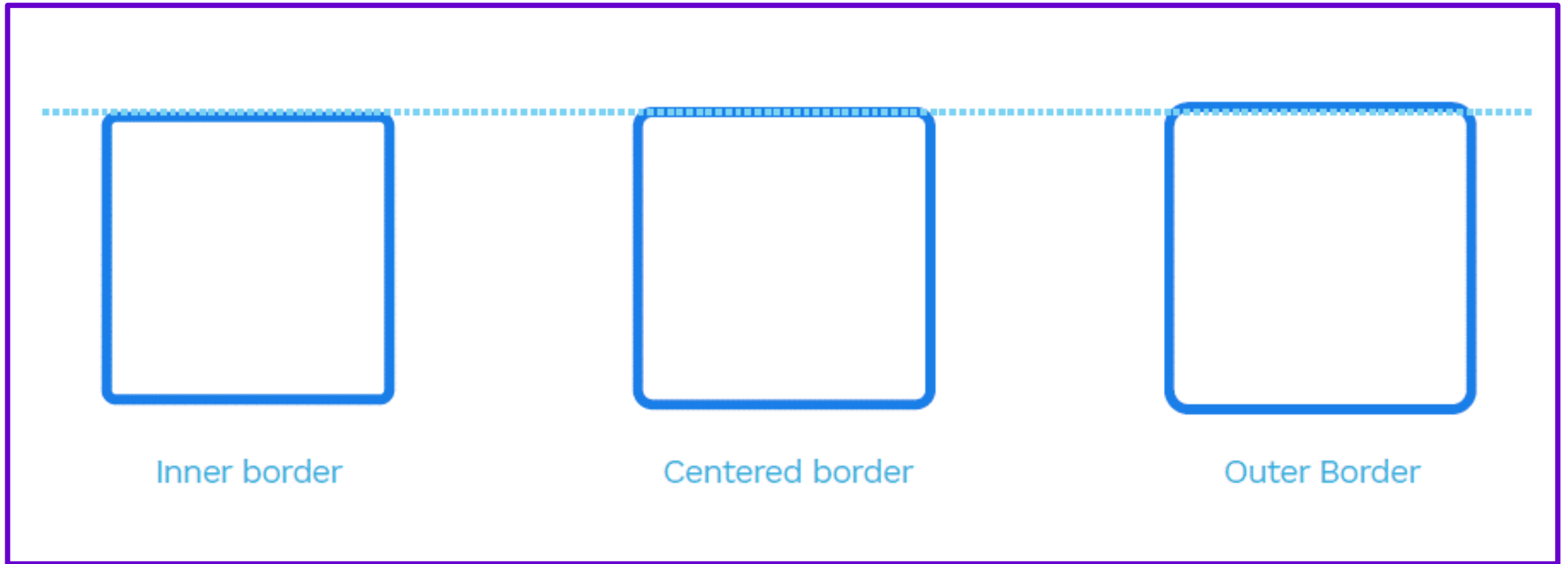


Fade-out
Gradient

BORDER



BORDER ...



BORDER ...



Dashed



Color



Gradient



2pt



10pt

BORDER ...



Flat, open end



Flat end



Rounded end



Straight joint

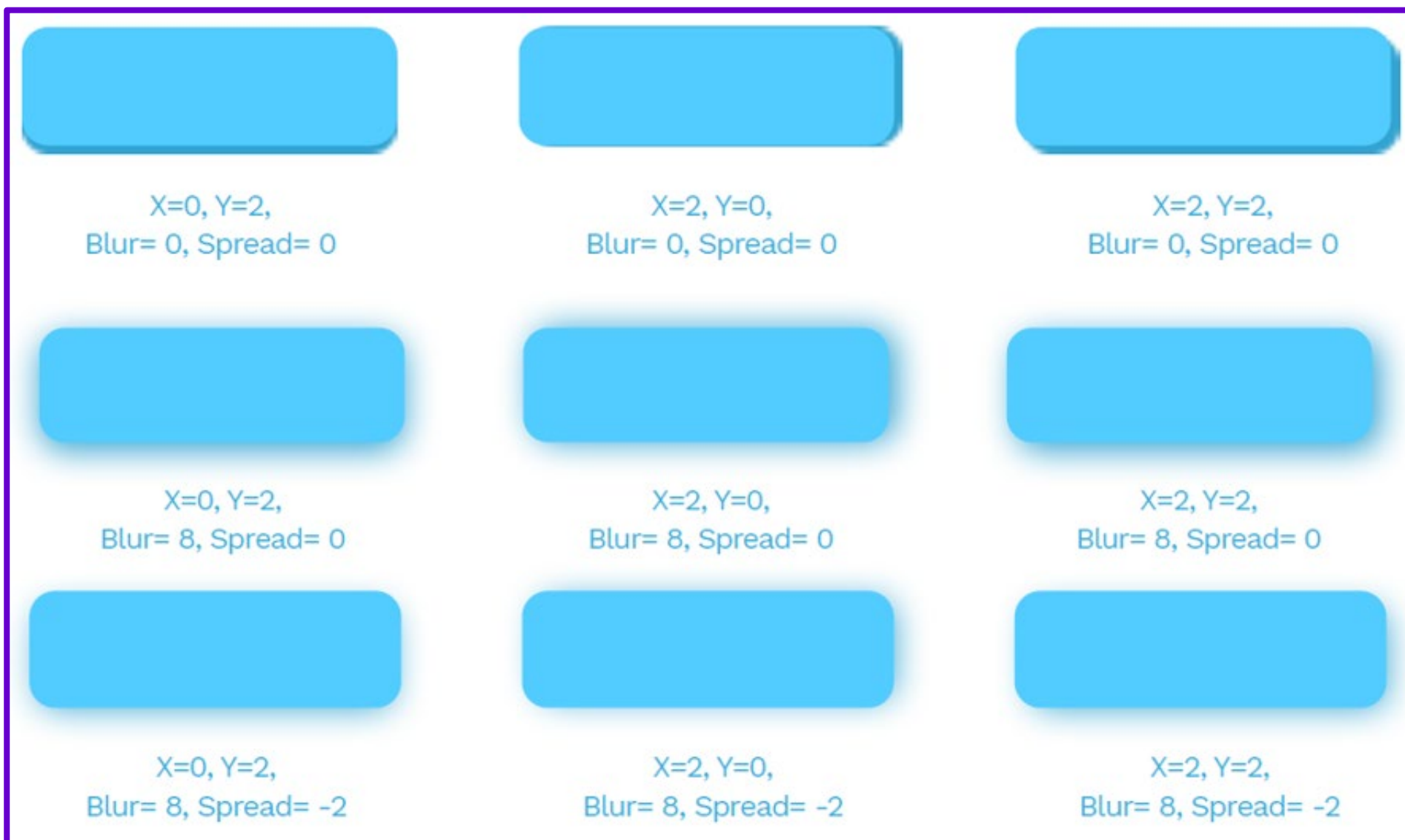


Rounded joint



Angled joint

DROP SHADOW



DROP SHADOW (Color)

The most important part of looking natural is avoiding pure black shadows and using a shadow derived from our primary color instead.



Our base color



Shadow color
Saturation -10
Brightness -20



Base color and
shadow combined.

DROP SHADOW (Color)

The most important part of looking natural is avoiding pure black shadows and using a shadow derived from our primary color instead.



INNER SHADOW

The only use-cases of this style are form inputs and extruded shapes in the Neumorphism method.



Outer shadow



Inner shadow

OUTER SHADOW

Modern UI's are using the shadows to build a visual hierarchy and layered structure of elements.



INNER SHADOW

An inner shadow makes us think the object has a hole in it, which breaks the expected layering structure and can be confusing.



GAUSSIAN BLUR

You can employ it into transitions between screens, or show a bit of realistic depth of field by selectively blurring the background



0



2p



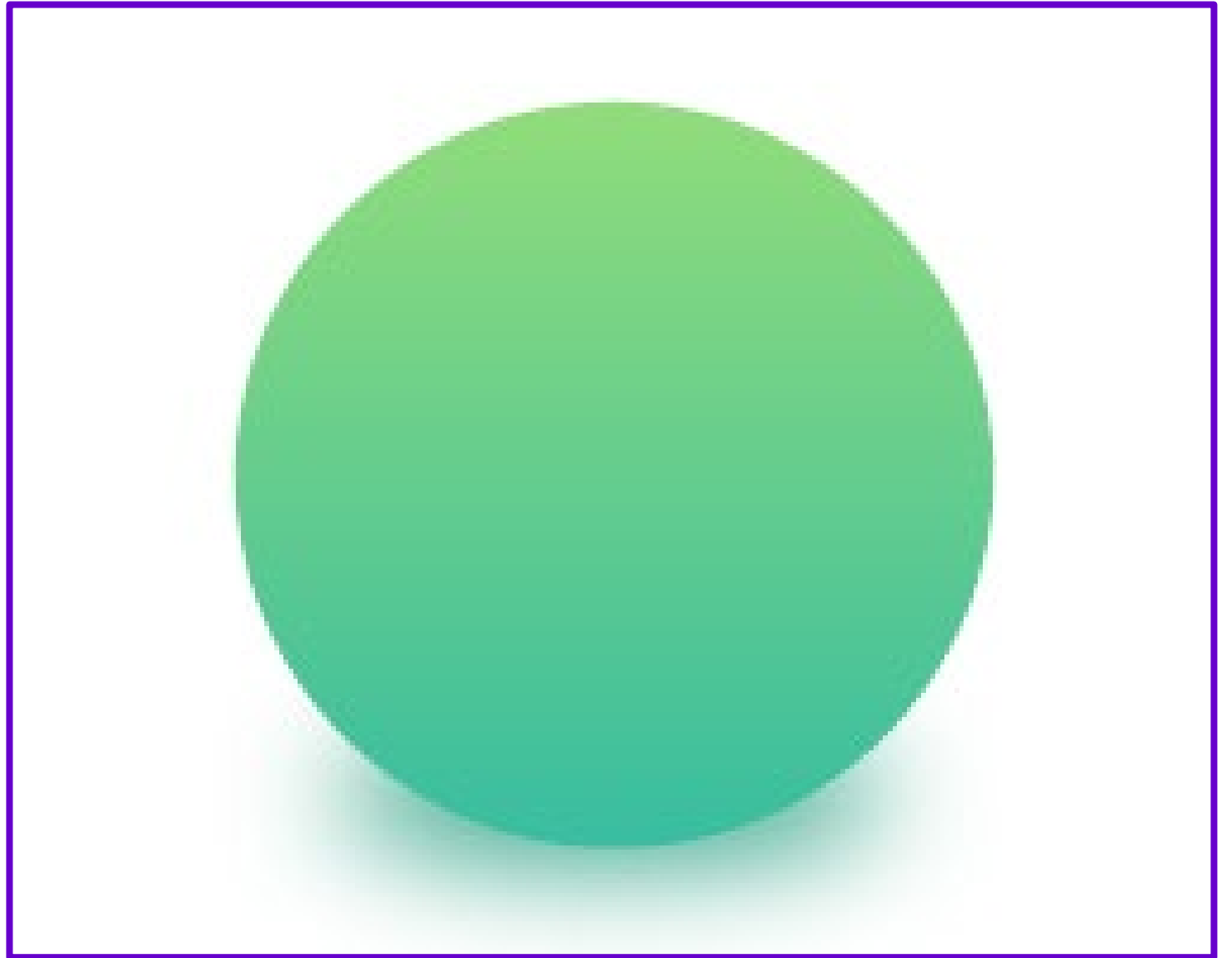
8p



16p

non-standard, point-
shadows under
objects

The easiest way to
achieve it is by
blurring an ellipse
and placing it under
the object casting
the shadow.



BACKGROUND BLUR

An object with this effect applied blurs everything under it.



MOTION BLUR

A motion blur simulates the movement of an object in a direction defined by the angle value



No blur



0° Angle



45° Angle



90° Angle

ZOOM BLUR

A zoom blur happens when the object becomes blurry from the inside out.



Blur value 0



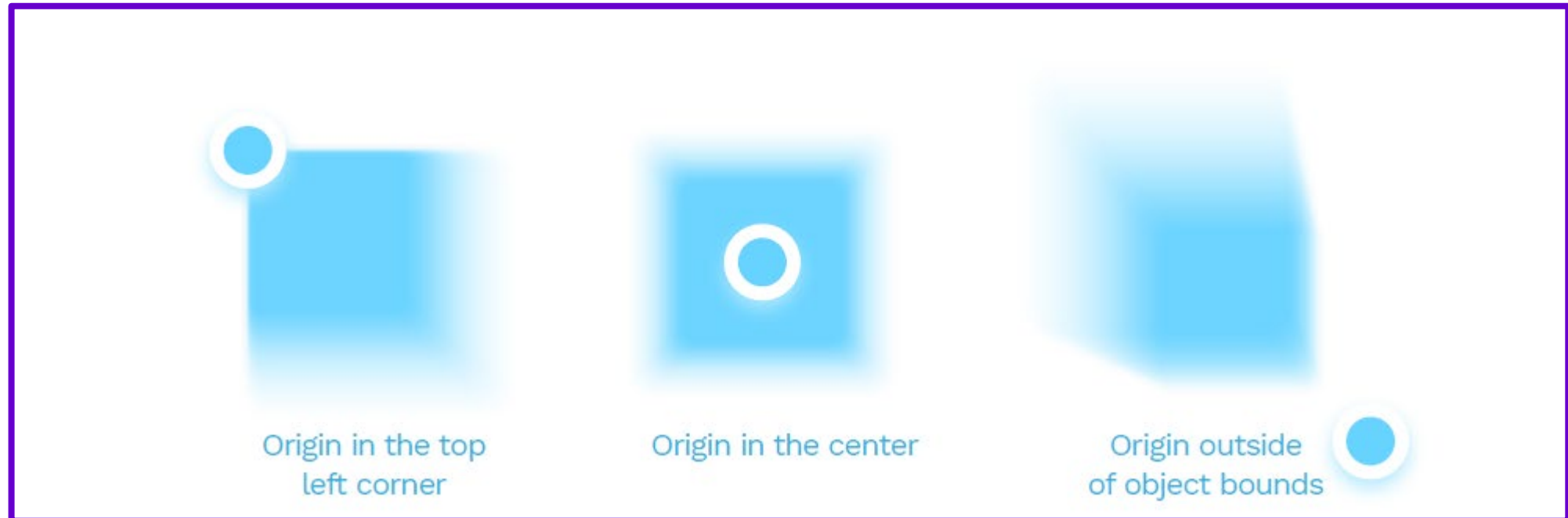
Blur value 25



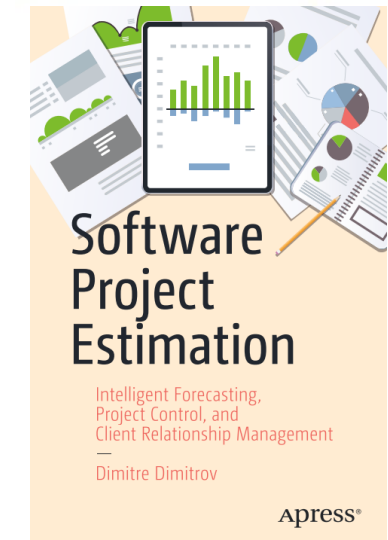
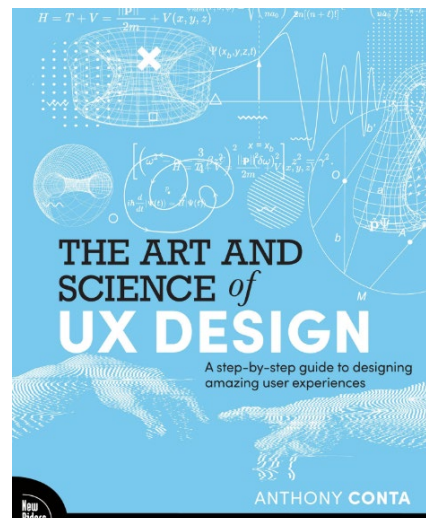
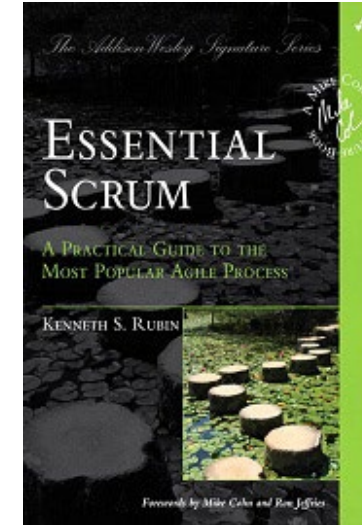
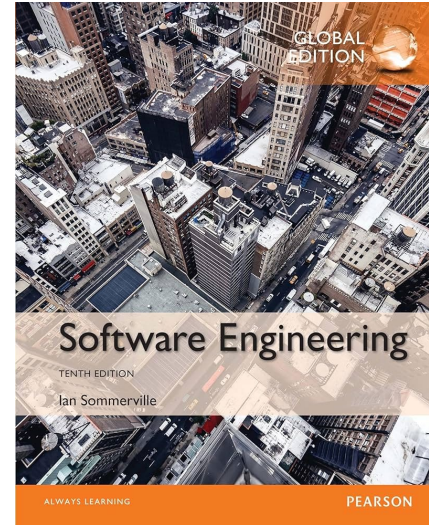
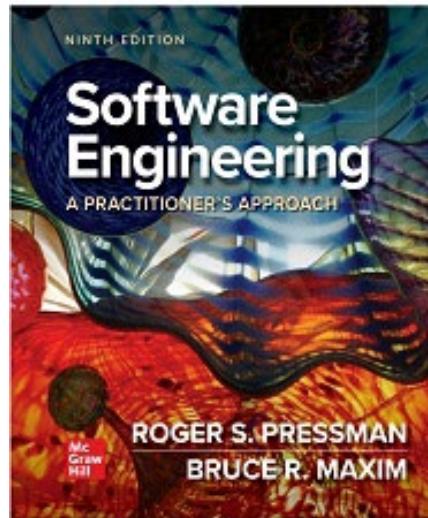
Blur value 50

ZOOM BLUR ...

In this particular blur type, you can also set the origin of the blur.



Course References



Course References

- ***Designing User Interfaces***, Michal Malewicz & Diana Malewice, 2020
- ***UI Design Styles: Trends and Design Patterns***, Michal Malewicz & Diana Malewice, 2020
- ***What UX Is Really About :Introducing a Mindset for Great Experiences***, Celia Hodent, CRC Press, 2022
- ***Lean UX: Designing Great Products with Agile Teams 3rd Edition***, Jeff Gothelf & Josh Seiden, O'Reilly, 2021
- ***Laws of UX: Using Psychology to Design Better Products & Services***, Jon Yablonski, O'Reilly, 2020
- ***Designing and Prototyping Interfaces with Figma***, Fabio Staiano, Packet Publishing, 2022

Course References

- [1] Green, M. D. (2016). Scrum: Novice to Ninja: Methods for Agile, Powerful Development, SitePoint.
- [2] Ockerman, S. and S. Reindl (2019). Mastering professional scrum: A practitioner's guide to overcoming challenges and maximizing the benefits of agility, Addison-Wesley Professional.
- [3] Martin, R. C. (2019). Clean Agile, Pearson Education.
- [4] Hall, G. M. (2017). Adaptive Code: Agile coding with design patterns and SOLID principles, Microsoft
- [5] سامانی‌پور، علی. (۲۰۱۸). آموزش اسکرام برای مدیریت چابک فرایند توسعه اپلیکیشن های وب و موبایل،
فرادرس